



USER INTERFACE DESIGN

23CSE25242LAB

MANUAL

SUBMITTED BY	
NAME	Sk.Tasmiya
ROLL NO	AV.SC.U4CSE25242
YEAR/SEM/SECTION	1 ST YEAR/SEM-2/CSE-C
CREDITS	03

SUBMITTED TO	DR.RAJ KUMAR BATCHU
DESIGNATION	ASST.PROFESSOR(SL.GRADE)
DEPARTMENT	SCHOOL OF COUMPUTING

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S.NO	TITTLE	PAGE .NO	REMARKS
1.	WEEK 1 : 1. Installation of VS Code tool. 2. Understanding basic html Structure. 3. Understanding basic html tags	4-11	
2.	creating a web page using HTML.	12-15	
3.	Create class time table using HTML elements	16-27	
4.	a) A college wants to collect student details through an online admission form. Design a registration form using only HTML tags (no CSS) with proper labels, table alignment , background colour , and borders. b) A company wants to design a webpage that displays two different forms side by side using iframes . One Iframes should display a Login Form and the other should display a Registration Forms .	28-34	
5.	a).create a card layout using tag.consider three cards and a button for each card ,using HTML and CSS, b). Create a mini website using iframe with navigation (Home, About, Contact) that display content in the same page. c). Design an amrita login page showing different campuses on the homepage. On clicking each campus, display campus details on the same page using HTML and CSS to design stylish webpage.	35-49	
6.	Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. Based on the category we select corresponding products should be displayed on the sidebar	50-57	

7.	To create a new repository in git	57-59	
8.	A).Create a webpage that display a square that continuously rotates 360 degrees using css. B).Create a ball that moves up and down continuously using css animation	60-65	
9.	a)create a button for each program based on the button clicked the corresponding program has to be excuted and should display the output b)the logics for each button includes finding an even number, odd number, prime or not ,factorial of a number , armstrong number use appropriate designs to display the output	65-72	
10.	A).write a java script conditional statement to find the sign of a product of three numbers using functions,display an alert box with specified sign. B).write a javascript conditional statement to sort three numbers using functions,display an alert box to show the result C).write a javascript conditional statement to find the largest of five numbers using functions and onclick event, display an alert box to show the result	72-84	
11.	1. A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters. 2. A registration form should validate email format before submission. If invalid, the form should not be submitted. 3. A system should check password strength while typing and display whether it is weak or strong. 4. A form should validate a mobile number when the user leaves the input field. The number must be exactly 10 digits. 5. Design an HTML page, such that the registration form must validate	85-94	
12.	Explain the setup process of electron application in your desktop .	95-96	

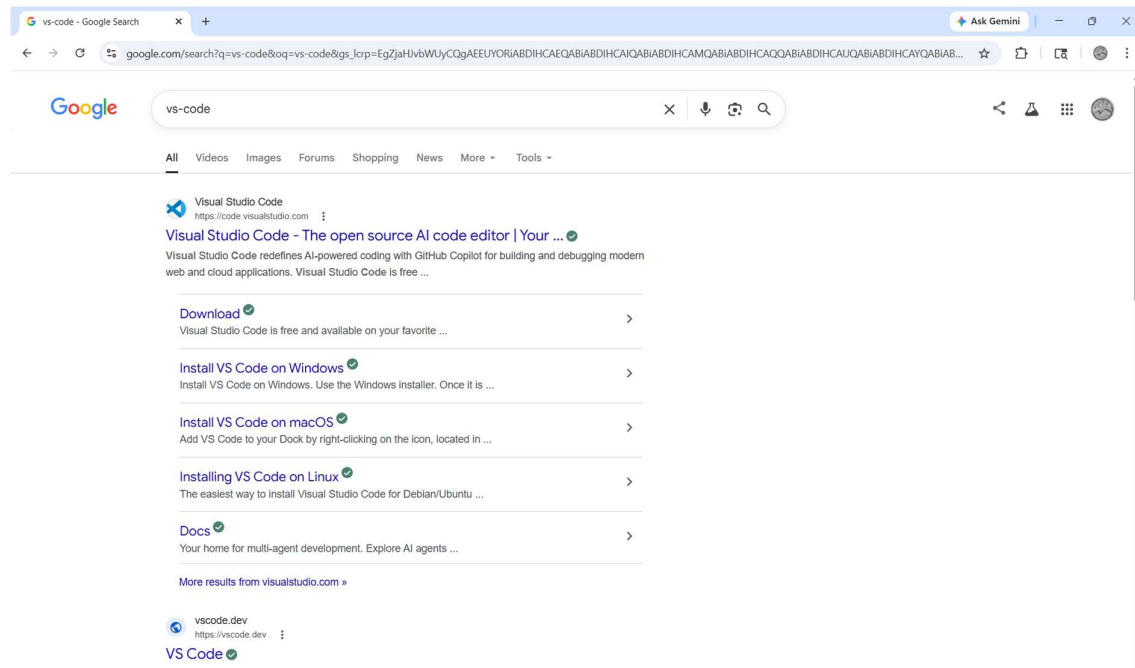
WEEK 1

1.AIM : steps to install VS COD

PROCEDURE:

STEP 1:open web-browser and click vs-code

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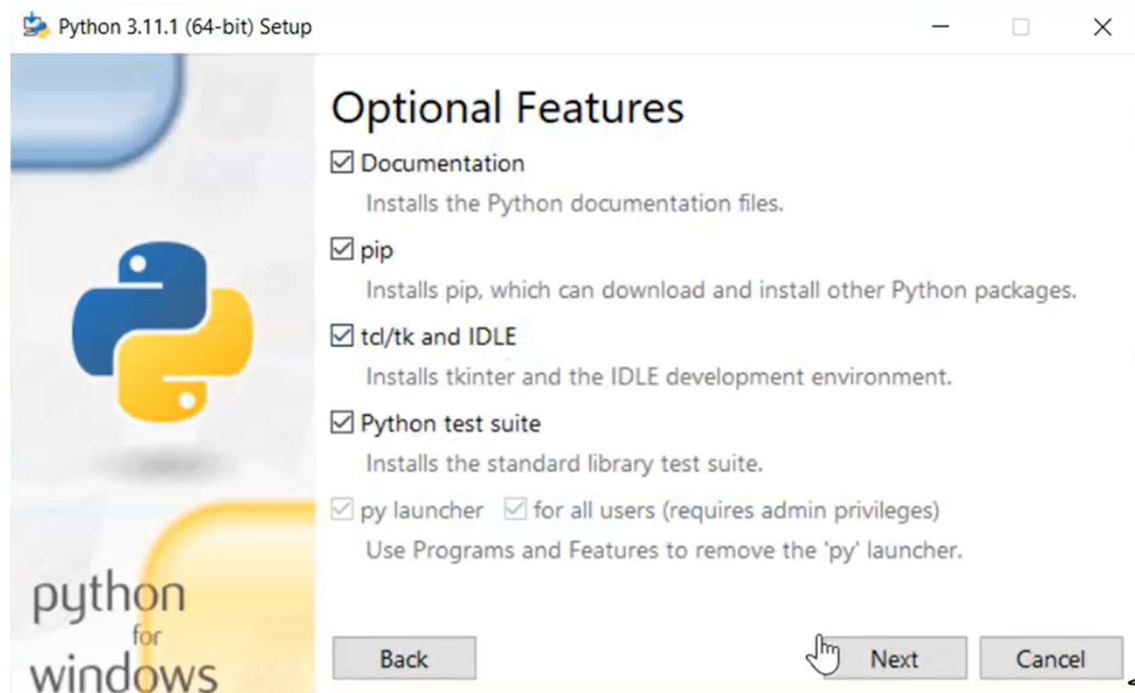
STEP 2:

The screenshot shows the Python.org website and the Python 3.11.1 (64-bit) Setup window. The website is the Python.org homepage, featuring the Python logo, navigation links (About, Downloads, Documentation, Community, Success Stories, News, Events), and a prominent "Download the latest version for Windows" button. Below this, there are links for "Download Python 3.11.1" and information about other operating systems (Windows, Linux/UNIX, macOS, Other). A banner for the "Support Python in 2021 End of year fundraiser and membership drive" is also visible. The "Active Python Releases" section shows a table with the following data:

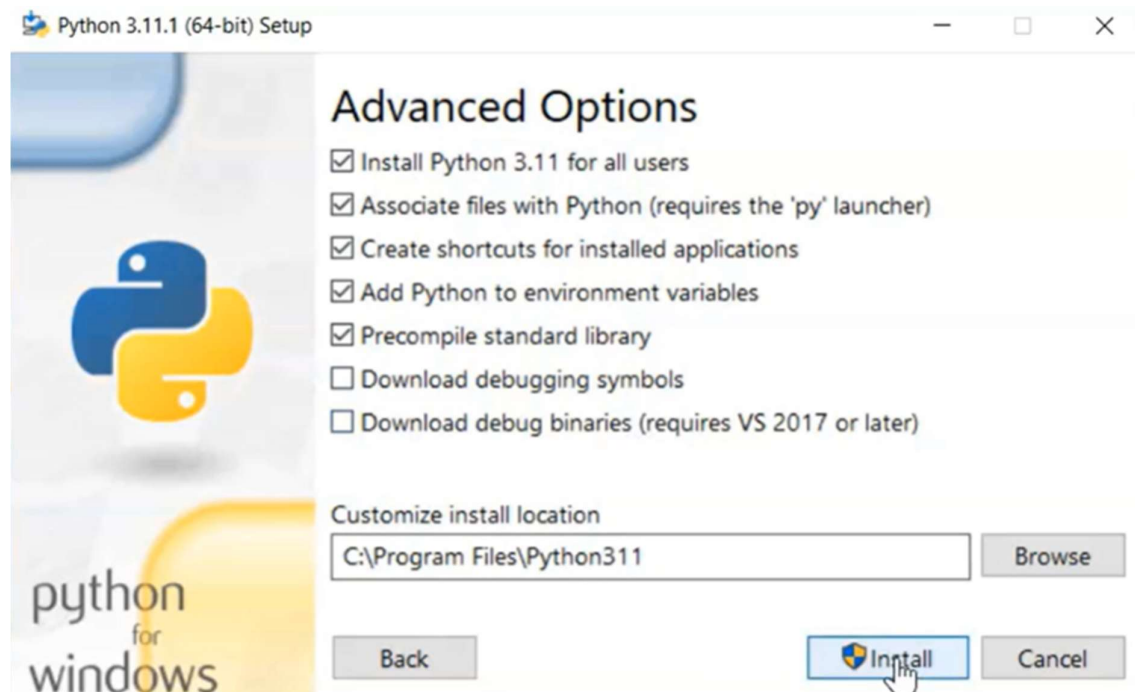
Python version	Maintenance status	First released	End of support	Release schedule
3.11	bugfix	2022-10-24	2027-10	PEP 664

The Python 3.11.1 (64-bit) Setup window is open, showing the "Install Python 3.11.1 (64-bit)" dialog. It offers two options: "Install Now" and "Customize installation". The "Install Now" option is selected, showing the installation path: `C:\Users\jaspr\AppData\Local\Programs\Python\Python311`. It also lists the included features: "Includes IDLE, pip and documentation" and "Creates shortcuts and file associations". The "Customize installation" option is also available, with the instruction "Choose location and features". At the bottom, there are two checked checkboxes: "Use admin privileges when installing py.exe" and "Add python.exe to PATH". A "Cancel" button is located in the bottom right corner.

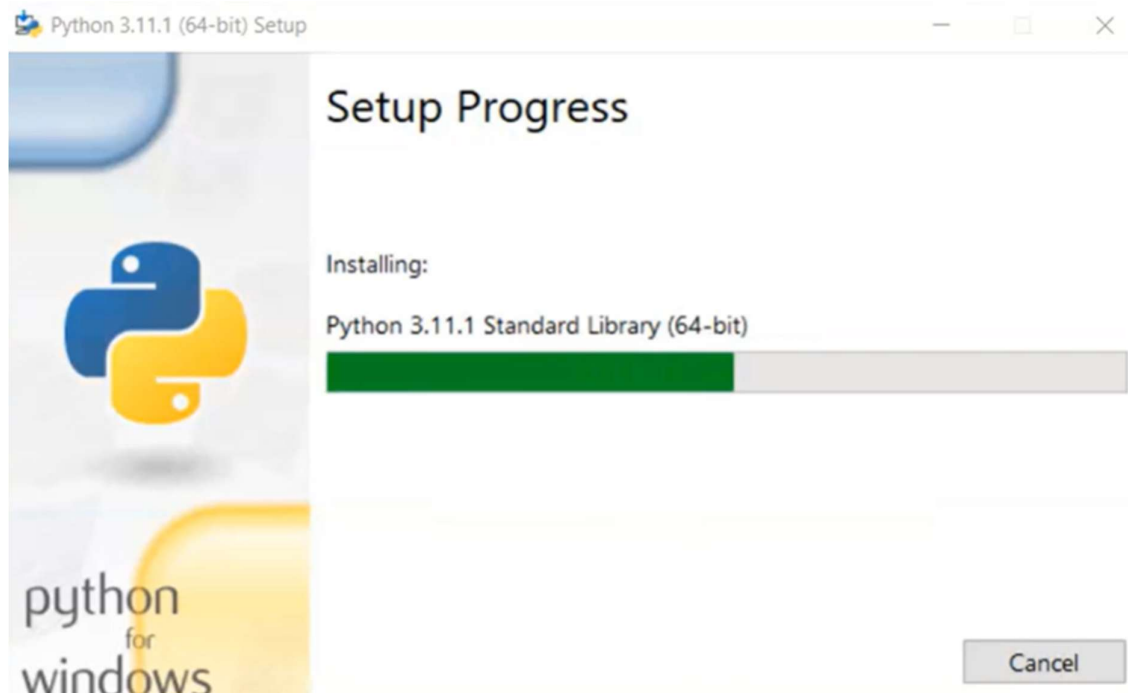
STEP 3:



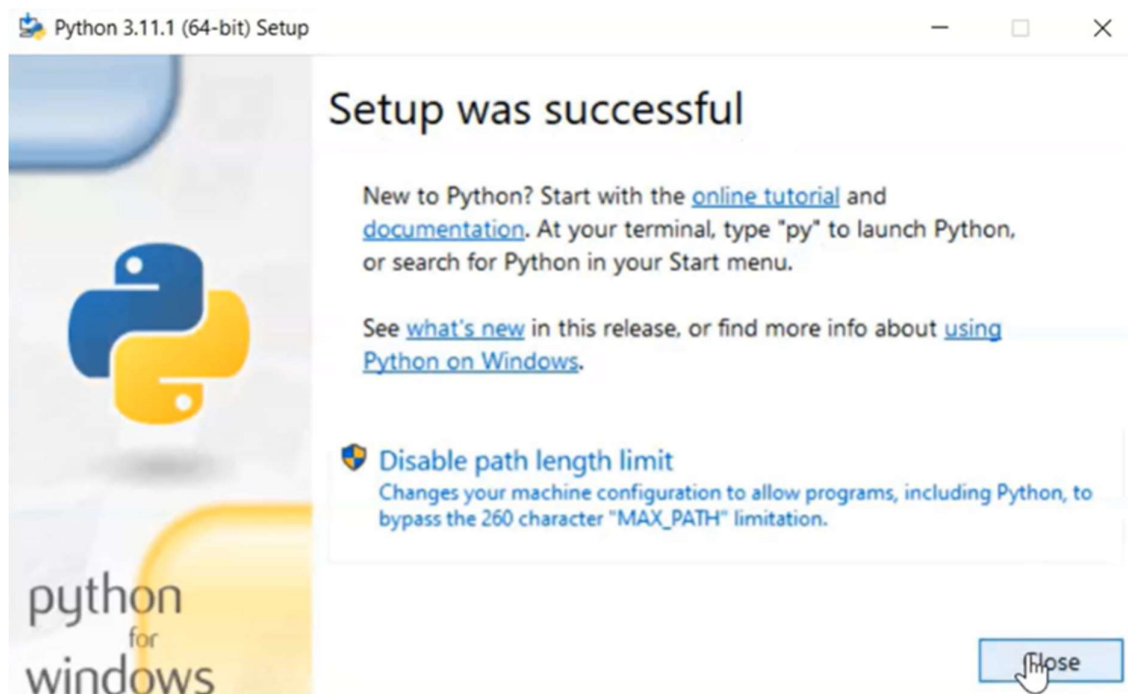
STEP 4:



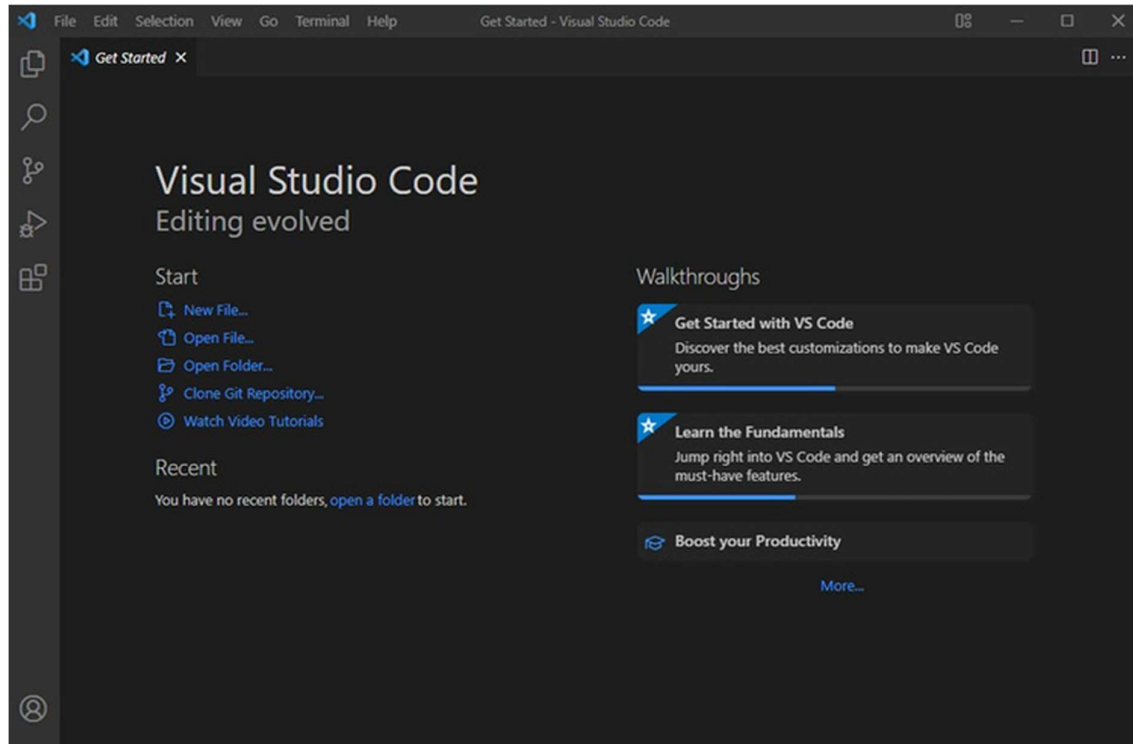
STEP 5:



STEP 6:



STEP 7:



2.AIM: TO LEARN BASICS OF HTML(basic.html):

Program:

UID-25242-TASMIYA

```
<!DOCTYPE html>
```

```
<html>
```

```
  <head>
```

```
    <title>firstwebpage</title>/
```

```
  </head>
```

```
  <body>
```

```
    <h1>happy to learn HTML</h1>
```

```
    <p>this is my first sample code</p>
```

```
  </body>
```

```
</html>
```

Output:



Explanation of the tags:

1. HTML stands for Hyper Text Markup Language.
 2. HTML is used to create and structure web pages.
- It tells the web browser what to display and how to display it
3. **<DOCTYPE HTML>**: The **<DOCTYPE html>** element tells the browser, this is an HTML5 document
 4. **<html>**: The element is the root element of an HTML page.
 5. **<head>**: The element contains meta information about the HTML page.
 6. **<title>**: The **<title>** element specifies a title for the HTML page.
 7. **<body>**: The **<body>** element contains all visible content like text, images, links, tables, etc.
 8. **<h3>**: The **<h3>** element defines a large heading.
 9. **<p>**: The **<p>** element defines a paragraph.

3. Aim :

understanding basic html tags.

Tags:- **<p>**; **<h1>**; ****; **<color>**

Program:

UID25242-Sk.tasmiya

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

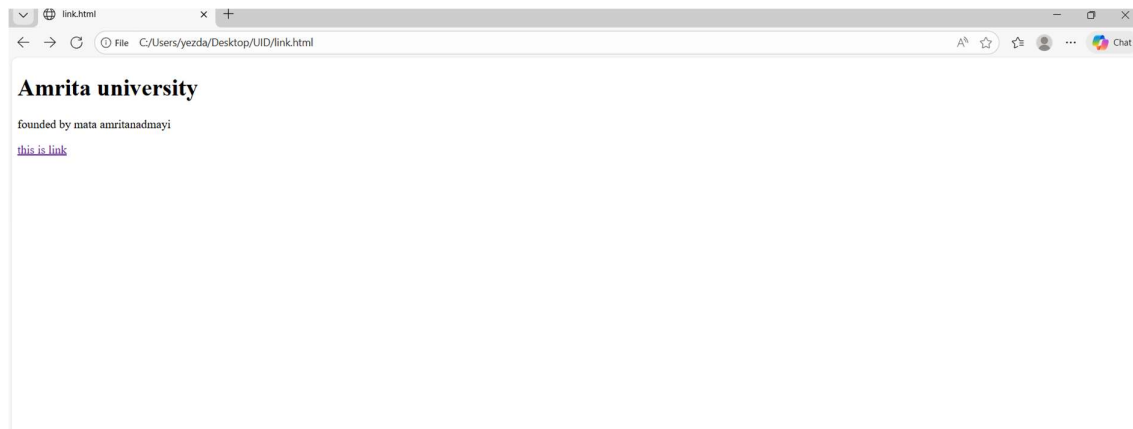
```
<h2>Amrita university</h2>
```

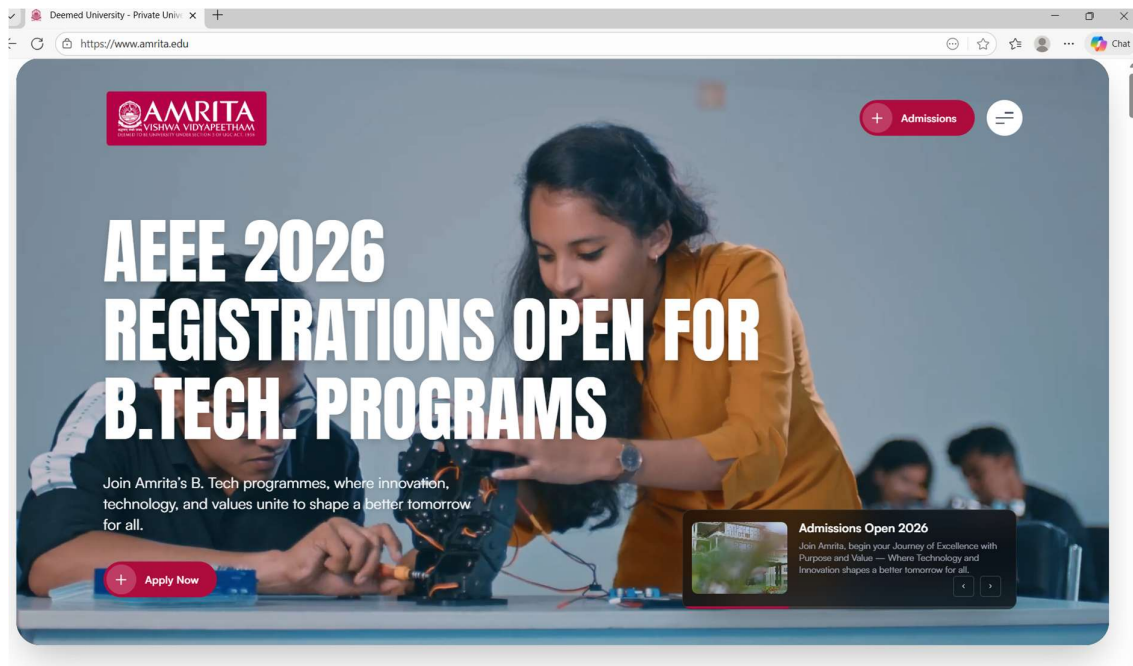
```
<p>founded<br> by mata amritanandamayi</p>
```

```
<h3>style attribute</h3>
```

```
<p style="color:red;">this is a red paragraph</p>
<a href="https://www.amrita.edu/">this is link</a>
<br><br>
<imgsrc="https://upload.wikimedia.org/wikipedia/commons
thumb/4/4e/Mata_Amritanandamayi_Devi.jpg/300px-Mata_Amritanandamayi_Devi.jpg">
</body>
</html>
```

Output:





Explanation of the tags:

⟨br⟩: The HTML element defines a line break. it is an empty tag with no end tag

Style: The HTML style attribute is used to add styles to an element such as color, font, size and more

⟨img⟩: the ⟨img⟩ tag is a fundamental HTML element used to embed an image into a web page

⟨a⟩ : The HTML tag ,which stands for “anchor”, is used to define a hyperlink

Week2

AIM:

creating a web page using HTML:

⟨!DOCTYPE html⟩

⟨html⟩

```
<body style="
background-image:url('Amravati-02.jpg');
background-repeat:no-repeat;
background-size:cover;
height:100vh;
width:100vw;
">
```

```
<h1 style="
color:maroon;
text-align:center;
border:4px double darkorange;
padding:8px;
background-color:rgba(255,255,255,0.6);
">
```

AMRITA VISHWAVIDYAPEETHAM

```
</h1>
```

```
<p style="
Text-align:center;
Font-family:Verdana, Geneva, Tahoma, sans-serif;
font-size:16px;
color:darkblue;
background-color:rgba(255,255,255,0.5);
width:85%;
margin:auto;
padding:10px;
">
```

Mata Amritanandamayi Devi (born Sudhamani Idamannel on September 27, 1953),
popularly known as Amma, the “hugging saint”, is a world-renowned
Indian spiritual leader and humanitarian from Kerala.

She is admired globally for spreading compassion and love
to reduce emotional and physical suffering.

<table
align="center"
border="2"
cellpadding="12"
cellspacing="6"
style="
background-image:url('Amma.jpg');
background-repeat:no-repeat;
background-size:cover;
border-style:ridge;
border-color:brown;
>

<caption style="
background-color:lightseagreen;
font-weight:bold;
padding:6px;
>

AMRITA VISHWAVIDYAPEETHAM BRANCHES

</caption>

<tr style="background-color:steelblue; color:white;">
<th>S.NO</th>
<th>CAMPUS</th>
<th>LOCATION</th>

```
<th>CONTACT NO</th>
<th colspan="2">CONTACT PERSON</th>
</tr>

<tr>
<td style="background-color:plum; text-align:center;">1</td>
<td style="background-color:lightgreen;">Amaravati Campus</td>
<td style="background-color:khaki;">Kuragallu</td>
<td rowspan="2" style="text-align:center;">+91-9865784585</td>
<td>Rajkumar</td>
<td>Batchu</td>
</tr>

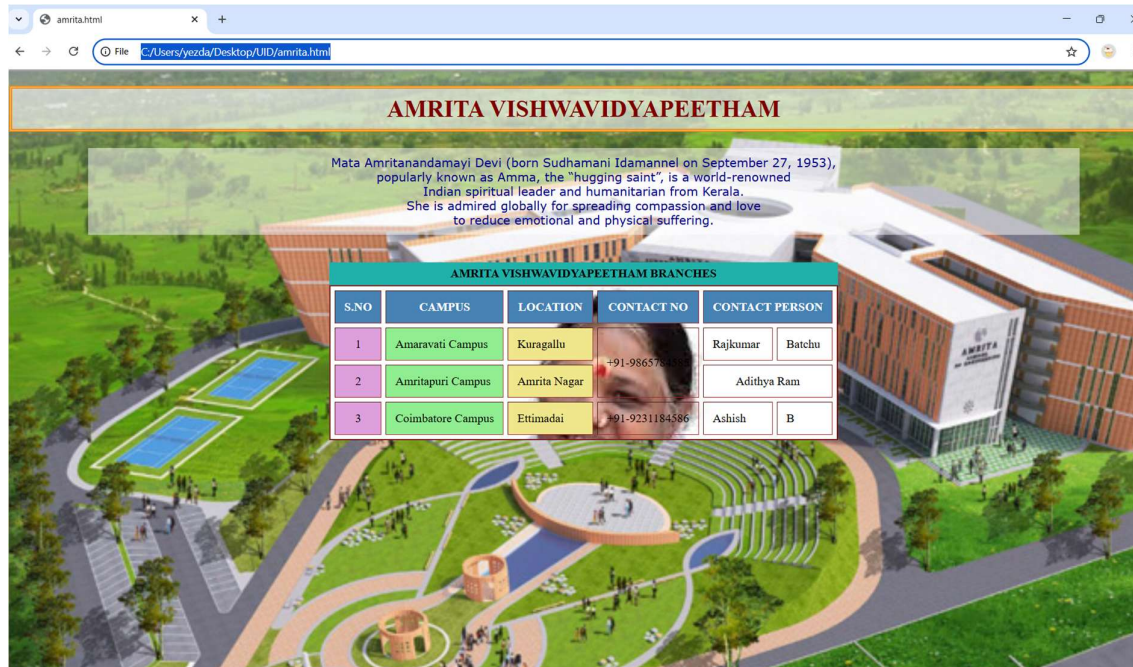
<tr>
<td style="background-color:plum; text-align:center;">2</td>
<td style="background-color:lightgreen;">Amritapuri Campus</td>
<td style="background-color:khaki;">Amrita Nagar</td>
<td colspan="2" style="text-align:center;">Adithya Ram</td>
</tr>

<tr>
<td style="background-color:plum; text-align:center;">3</td>
<td style="background-color:lightgreen;">Coimbatore Campus</td>
<td style="background-color:khaki;">Ettimadai</td>
<td style="text-align:center;">+91-9231184586</td>
<td>Ashish</td>
<td>B</td>
</tr>

</table>
```


</body>

</html>



Explanation of tags:

 : This gives a break . It means that we will go to the next line

<h1>,<h2>,<h3> : These are the opening tags of headings respectively .

And </h1>,</h2>,</h3> are the closing tags of those headings.

<p style : This is the style attribute . It gives background colours, highlighting the text with colour , font size, and more

WEEK 3

AIM:Create class time table using HTML elements:

Program:

<!DOCTYPE html>

<html>

<body>

```

<table border="2px">
  <caption style="text-align: center; background-color: yellow;" border="2px">MASTER
  TIME TABLE CSE 2-SEMESTER C-SECTION</caption>
  <tr style="text-align: center;">
    <td colspan="4">Dept. CSE</td>
    <td rowspan="3" colspan="8">Class Room Venue: 512</td>
  </tr>
  <tr style="text-align: center;">
    <td colspan="4">Class: B. Tech. CSE-C</td>
  </tr>
  <tr style="text-align: center;">
    <td colspan="4">Semester-II</td>
  </tr>
  <tr style="text-align: center;">
    <td colspan="4">Class Advisors: <br>
      1. Name: Mr. Budati Jaya Lakshminarayana<br>
      Email : bj_lakshminarayana@av.amrita.edu<br>
      Mobile Number :+91-7989795768
    </td>
    <td colspan="8">2. Name: Dr. Arun Kumar S<br>
      Email : s_arunkumar@av.amrita.edu<br>
      Mobile Number : +91-8940857202
    </td>
  </tr>
  <tr style="text-align: center;">
    <td rowspan="2">Time/ Day</td>
    <td>Slot 1</td>
    <td>Slot 2</td>
    <td rowspan="2">10.35am-
      10.45am</td>
    <td>Slot 3</td>

```

```

<td>Slot 4</td>
<td>Slot 5</td>
<td rowspan="2">01.15pm
-02.05pm</td>
<td>Slot 6</td>
<td rowspan="2">01.15pm
-02.05pm</td>
<td>Slot 7</td>
<td>Slot 8</td>
</tr>
<tr style="text-align: center;">
<td>08.55-09.45am</td>
<td> 09.45-10.35am</td>
<td>10.45-11.35am</td>
<td> 11.35am-12.25pm</td>
<td> 12.25am-01.15pm</td>
<td> 02.10-03.00pm</td>
<td> 03.10-04.00pm</td>
<td> 04.00-04.50pm</td>
</tr>
<tr style="text-align: center;">
<td>Monday</td>
<td colspan="2">23MAT116 Discrete Mathematics_lab2</td>
<td style="background-color: yellow;writing-mode: sideways-lr;" rowspan="5">Short
Break</td>
<td> 23PHY115 Modern Physics</td>
<td>23CSE111Object Oriented Programming</td>
<td>Library</td>
<td style="background-color: yellow;writing-mode: sideways-lr;" rowspan="5">Lunch
Break</td>
<td>Library</td>

```

```

        <td style="background-color: yellow;writing-mode: sideways-lr;" rowspan="5">Short
Break</td>

```

```

        <td colspan="2">Library/Sports</td>

```

```

    </tr>

```

```

    <tr style="text-align: center;">

```

```

        <td>Tuesday</td>

```

```

        <td colspan="2">23CSE113 User Interface Design LAB(BYOD)</td>

```

```

        <td>23MAT117 Linear Algebra</td>

```

```

        <td>23MAT116 Discrete Mathematics</td>

```

```

        <td>COUNSELLING</td>

```

```

        <td>Library</td>

```

```

        <td colspan="2">22ADM111 Glimpses of Glorious India</td>

```

```

    </tr>

```

```

    <tr style="text-align: center;">

```

```

        <td>Wednesday</td>

```

```

        <td>23PHY115 Modern Physics</td>

```

```

        <td>23CSE113 User Interface Design</td>

```

```

        <td colspan="2">23MAT117 Linear Algebra_LAB(BYOD)</td>

```

```

        <td>COUNSELLING</td>

```

```

        <td>Library</td>

```

```

        <td colspan="2">Library/Sports</td>

```

```

    </tr>

```

```

    <tr style="text-align: center;">

```

```

        <td>Thursday</td>

```

```

        <td>23CSE111Object Oriented Programming</td>

```

```

        <td>23MAT116 Discrete Mathematics</td>

```

```

        <td colspan="2">23CSE111Object Oriented Programming_Lab2</td>

```

```

        <td>23MAT117 Linear Algebra</td>

```

```

        <td>23MEE115 Manufacturing Practice LAB</td>

```

```

        <td colspan="2">23MEE115 Manufacturing Practice LAB</td>

```

```

    </tr>

```

```

<tr style="text-align: center;">
  <td>Friday</td>
  <td>23CSE111Object Oriented Programming</td>
  <td>23MAT116 Discrete Mathematics</td>
  <td>23PHY115 Modern Physics(T)</td>
  <td>23CSE113 User Interface Design</td>
  <td>23MAT117 Linear Algebra</td>
  <td>22ADM111Glimpses of Glorious India(P)</td>
  <td colspan="2">Library/Sports</td>
</tr>
</table>
<br>
<hr>
<br>
<table border="2px" style="border-collapse: collapse;margin: auto;" class="center"
height="300" width="850" >
<tr style="text-align: center;">
  <td>Course Code</td>
  <td width="200px">Course Name</td>
  <td width="50px">  </td>
  <td width="70px">L-T-P</td>
  <td width="70px">Credits</td>
  <td width="200px">Faculty Name</td>
  <td width="50px"></td>
  <td width="200px">Assisting Faculty</td>
</tr>
<tr>
  <td>23MAT116</td>
  <td>Discrete Mathematics</td>
  <td>  </td>
  <td style="text-align: center;">3-0-2</td>

```

```

<td style="text-align: center;">4</td>
<td> Dr.Srivivasa Rao Thota</td>
<td></td>
<td></td>
</tr>
<tr>
<td>23MAT117</td>
<td>Linear Algebra</td>
<td> </td>
<td style="text-align: center;">3-0-2 </td>
<td style="text-align: center;">4</td>
<td> Mr.Sapavat Bixapathi</td>
<td></td>
<td></td>
</tr>
<tr>
<td>23CSE111</td>
<td>Object Oriented Programming</td>
<td> </td>
<td style="text-align: center;">3-0-2</td>
<td style="text-align: center;">4</td>
<td>Dr.Satya Rajendra Singh R</td>
<td></td>
<td>Dr. Balaji TK</td>
</tr>
<tr>
<td>23PHY115</td>
<td>Modern Physics</td>
<td> </td>
<td style="text-align: center;">2-1-0</td>
<td style="text-align: center;">3</td>
<td>Dr.Soumya</td>

```

```

<td></td>
<td></td>
</tr>
<tr>
  <td>23CSE113</td>
  <td>User Interface Design</td>
  <td>  </td>
  <td style="text-align: center;">2-0-2</td>
  <td style="text-align: center;">3</td>
  <td> Dr.Rajkumar Batchu</td>
  <td></td>
  <td>Mr. Barge Bharadwaj</td>
</tr>
<tr>
  <td>23MEE115</td>
  <td>Manufacturing Practice </td>
  <td>  </td>
  <td style="text-align: center;">0-0-3</td>
  <td style="text-align: center;">1</td>
  <td>Dr. Sathies S</td>
  <td></td>
  <td></td>
</tr>
<tr>
  <td>22ADM111</td>
  <td>Glimpses of Glorious India</td>
  <td>  </td>
  <td style="text-align: center;">2-0-1</td>
  <td style="text-align: center;">2</td>
  <td>Dr. Siva Panuganti</td>
  <td></td>

```

```

        <td></td>
    </tr>
    <tr>
        <td></td>
        <td style="text-align: center;">Total (15L + 1Tut + 12P =28 hrs)</td>
        <td cellpadding="10"></td>
        <td style="text-align: center;">15-1-12</td>
        <td style="text-align: center;">21</td>
        <td></td>
        <td></td>
        <td></td>
    </tr>
</table>

<br>
<br>
<br>
<hr>
<br>
<br>
<br>

<h1 style="text-align: center;background-color: yellow;" ><b> <strong> SYLLABUS (SEM - 2)
</strong></b></h1>

<p><b>1.Discrete Mathematics (23MAT116)</b></p>

<p><b>2.Linear Algebra (23MAT117)</b></p>

<p><b>3.Object Oriented Programming (23CSE111)</b></p>

<p><b>4.Modern Physic (23PHY115)</b></p>

<p><b>5.User Interface Design (23CSE113)</b></p>

```


<p>6.Manufacturing Practice (23MEE115)</p>

<p>7.Glimpses of Glorious India (22ADM111)</p>

<hr>

<p style="text-align: justify;"> ABOUT YOURSELF
 MY NAME IS SHAIK TASMIYA
 SECTION : CSE C
 ROLL NO : AV.SC.U4CSE25242 </P>

</body>

</html>

OUPUT:

MASTER TIME TABLE CSE 2-SEMESTER C-SECTION											
Dept. CSE				Class Room Venue: 512							
Class: B. Tech. CSE-C											
Semester-II											
Class Advisors:											
1. Name: Mr. Budati Jaya Lakshminarayana Email : bj_lakshminarayana@av.amrita.edu Mobile Number : +91-7989795768				2. Name: Dr. Arun Kumar S Email : s_arunkumar@av.amrita.edu Mobile Number : +91-8940857202							
Time/ Day	Slot 1 08.55-09.45am	Slot 2 09.45-10.35am	10.35am-10.45am	Slot 3 10.45-11.35am	Slot 4 11.35am-12.25pm	Slot 5 12.25am-01.15pm	01.15pm-02.05pm	Slot 6 02.10-03.00pm	01.15pm-02.05pm	Slot 7 03.10-04.00pm	Slot 8 04.00-04.50pm
Monday	23MAT116 Discrete Mathematics_lab2		Short Break	23PHY115 Modern Physics	23CSE111 Object Oriented Programming	Library	Lunch Break	Library	Short Break	Library/Sports	
Tuesday	23CSE113 User Interface Design LAB(BYOD)			23MAT117 Linear Algebra	23MAT116 Discrete Mathematics	COUNSELLING		Library		22ADM111 Glimpses of Glorious India	
Wednesday	23PHY115 Modern Physics	23CSE113 User Interface Design		23MAT117 Linear Algebra LAB(BYOD)		COUNSELLING		Library		Library/Sports	
Thursday	23CSE111 Object Oriented Programming	23MAT116 Discrete Mathematics		23CSE111 Object Oriented Programming Lab2	23MAT117 Linear Algebra			23MEE115 Manufacturing Practice LAB		23MEE115 Manufacturing Practice LAB	
Friday	23CSE111 Object Oriented Programming	23MAT116 Discrete Mathematics		23PHY115 Modern Physics(T)	23CSE113 User Interface Design	23MAT117 Linear Algebra		22ADM111 Glimpses of Glorious India(P)		Library/Sports	

Course Code	Course Name	L-T-P	Credits	Faculty Name	Assisting Faculty
23MAT116	Discrete Mathematics	3-0-2	4	Dr.Srivivasa Rao Thota	
23MAT117	Linear Algebra	3-0-2	4	Mr.Sapavat Bixapathi	
23CSE111	Object Oriented Programming	3-0-2	4	Dr.Satya Rajendra Singh R	Dr. Balaji TK
23PHY115	Modern Physics	2-1-0	3	Dr.Sounya	
23CSE113	User Interface Design	2-0-2	3	Dr.Rajkumar Batchu	Mr. Barge Bharadwaj
23MEE115	Manufacturing Practice	0-0-3	1	Dr. Sathies S	
22ADM111	Glimpses of Glorious India	2-0-1	2	Dr. Siva Panuganti	
	Total (15L + 1Tut +12P =28 hrs)	15-1-12	21		

CSE SEM2.html x +

File C:/Users/yezda/Desktop/UID/CSE%20SEM2.html

SYLLABUS (SEM - 2)

1. Discrete Mathematics (23MAT116)

Syllabus

Unit 1
Logic, Mathematical Reasoning and Counting: Logic, Propositional Equivalence, Predicate and Quantifiers, Theorem Proving, Recursive Definitions, Recursive Algorithms, Basics of Counting, Pigeonhole Principle, Permutation and Combinations.

Unit 2
Relations and Their Properties: Representing Relations, Closure of Relations, Partial Ordering, Equivalence Relations and partitions.
Advanced Counting Techniques and Relations: Recurrence Relations, Solving Recurrence Relations, Generating Functions, Solutions of Homogeneous Recurrence Relations, Divide and Conquer Relations, Inclusion-Exclusion.

Unit 3
Graph Theory: Graphs and Sub graphs, isomorphism, matrices associated with graphs, degrees, walks, connected graphs, shortest path algorithm. Euler and Hamilton Graphs: Euler graphs, Euler's theorem. Fleury's algorithm for Eulerian trails. Hamilton cycles, Chinese-postman problem, approximate solutions of traveling salesman problem. Closest neighbour algorithm.

CSE SEM2.html x +

File C:/Users/yezda/Desktop/UID/CSE%20SEM2.html

2. Linear Algebra (23MAT117)

Syllabus

Unit 1
 Vector Spaces: Vector spaces - Sub spaces - Linear independence - Basis - Dimension; Inner Product Spaces: Inner products - Orthogonality - Orthogonal basis - Gram Schmidt Process - Change of basis - Orthogonal complements - Projection on subspace - Least Square Principle. QR- Decomposition.

Unit 2
 Linear Transformations: Linear transformation - Relation between matrices and linear transformations - Kernel and range of a linear transformation - Change of basis - Nilpotent transformations. Symmetric and Skew Symmetric Matrices, Adjoint and Hermitian Adjoint of a Matrix, Hermitian, Unitary and Normal Transformations, Self-Adjoint and Normal Transformations.

Unit 3
 Eigen values and Eigen vectors: Eigen Values and Eigen Vectors, Diagonalization, Orthogonal Diagonalization, Quadratic Forms, Diagonalizing Quadratic Forms, Conic Sections. Similarity of linear transformations - Diagonalization and its applications - Jordan form and rational canonical form. Case Studies: Applications on least square and image transformations.

applications - Jordan form and rational canonical form. Case Studies: Applications on least square and image transformations.

CSE SEM2.html x +

File C:/Users/yezda/Desktop/UID/CSE%20SEM2.html

3. Object Oriented Programming (23CSE111)

Syllabus

Unit 1
 Structured to Object Oriented Approach by Examples - Object Oriented languages - Properties of Object Oriented system - UML and Object Oriented Software Development - Use case diagrams and documents as a functional model - Identifying Objects and classes - Representation of Objects and its state by Object Diagram - Simple Class using class diagram - Encapsulation - Data Hiding - Reading and Writing Objects - Class Level and Instance Level Attributes and Methods- JIVE environment for debugging.

Unit 2
 Aggregation and Composition using Class Diagram - Generalization using Class Diagram - Inheritance - Constructor and Over Riding - Visibility - Attribute - Parameter - Package - Local and Global - Polymorphism - Overloading - Abstract Classes and Interfaces.

Unit 3
 Exception Handling - Inner Classes - Wrapper classes - String - and String Builder classes - Number - Math - Random - Array methods - File Streams - Serialization - Generics - Collection framework - Comparator and Comparable - Vector and ArrayList - Iterator and Iterable.



4.Modern Physics (23PHY115)

Syllabus

Unit 1

Origin of quantum theory of radiation: Black body radiation, photo-electric effect, Compton Effect – pair production and annihilation, De-Broglie hypothesis, description of waves and wave packets, group velocities. Evidence for wave nature of particles: Davisson-Germer experiment, Heisenberg uncertainty principle.

Unit 2

Atomic structure: Historical Development of atomic structures: Thomson's Model, Rutherford's Model: Scattering formula and its predictions, Atomic spectra - Bohr's Model, Sommerfield's Model, The correspondence principle, nuclear motion, and atomic excitation, Application: Lasers.

Unit 3

Quantum mechanics: Wave function, Probability density, expectation values - Schrodinger equation – time dependent and independent, Linearity and superposition, expectation values, operators, Eigen functions and Eigen values

Unit 4

Application of 1D Schrodinger Wave equation: Free particle, Particle in a box, Finite potential well, Tunnel effect, Harmonic oscillator.

Unit 5

Intro to Quantum computing- Q bits- II Quantum correlations: Bell inequalities and entanglement, Schmidt decomposition, super dense coding, teleportation. Module



5.User Interface Design (23CSE113)

Syllabus

Unit 1

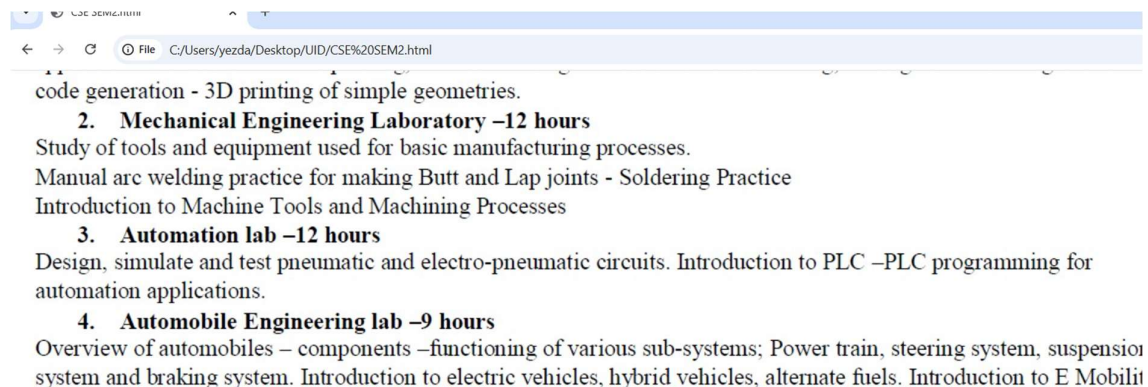
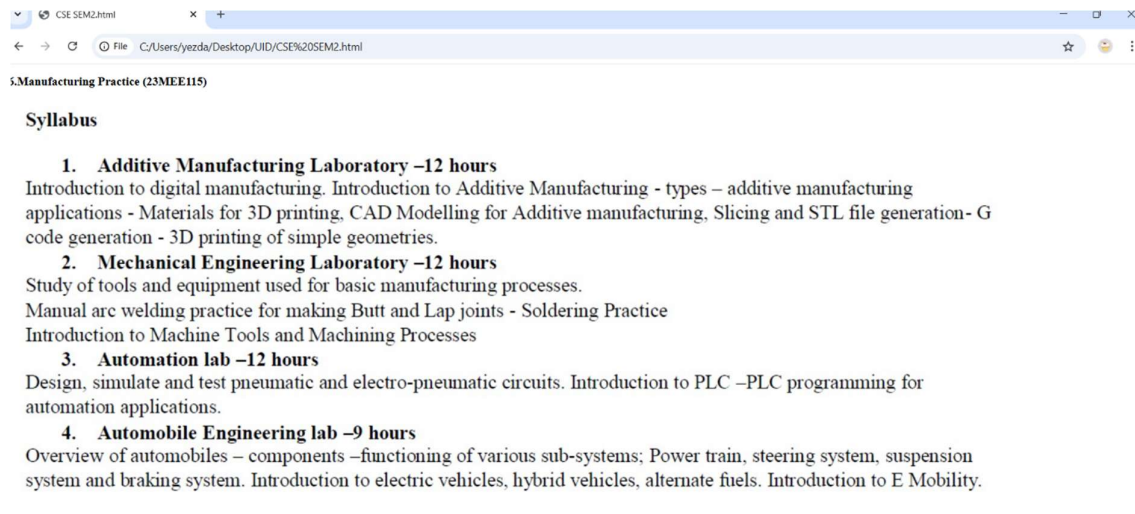
Introduction to Internet and Web design – Working of Web – Roles of Front-end, Back-end and Full stack development – Web Server – Internet protocols – Web Hosting – HTML – HTML Tags – Create a simple Web Page – Marking up Text – Adding Links – Adding Images – SVG – Table Markup – Embedded Media - Web Based Forms – HTML Forms - CSS for Presentation - Basic Style Sheet - A CSS Style Primer - Using Style Classes - Using Style IDs - Formatting Text - Advanced Typography with CSS3 - Working with Margins, Padding, Alignment, and Floating. Responsive web design, Introduction to version control systems.

Unit 2

CSS Box Model and Positioning - Creating Layouts Using Modern CSS - Backgrounds and Borders - CSS Transformations and Transitions - Animating with CSS and the Canvas - Dynamic Web Pages – Web Scripting - JavaScript programming for Web Applications – Including JavaScript in HTML - HTML5 Form Controls and Validation - Document Object Model - DOM manipulation and events - Event Handlers

Unit 3

Overview of UI Frameworks – User Interface Design Concepts in Web Development – Accessibility - Introduction to Electron - Overview of Electron - Setting up an Electron project - Creating a basic desktop application with Electron - Object communication - Introduction to other popular frameworks and libraries (e.g., JavaFX, WPF, GTK).



7.Glimpses of Glorious India (22ADM111)

Syllabus

Face the Brutes, Role of Women in India, Acharya Chanakya, God and Iswara, Bhagavad Gita: From Soldier to Samsarin to Sadhaka, Lessons of Yoga from Bhagavad Gita, Indian Soft powers, Preserving Nature through Faith, Ancient Indian Cultures (Class Activity), Practical Vedanta, To the World from India, Indian Approach to Science.

ABOUT YOURSELF
MY NAME IS SHAIK TASMIYA
SECTION : CSE C
ROLL NO : AV.SC.U4CSE25242

Explanation of tags:

1. **<table>**:element is used to create a table in a web page. It organizes data into rows and columns.
2. **Width**: The width attribute specifies the width of the table.
3. **border**: The border attribute defines the thickness of the table border.
4. **cellpadding**: The cellpadding attribute specifies the space between the cell content and its border.
5. **cellspacing**: The cellspacing attribute specifies the space between table cells.
6. **<caption>**:this element is used to give a title or heading to a table.it appears above the table by default.
7. **<tr>**:this element defines a table row inside a table.
8. **<th>**:this element defines a header cell in a table.the text inside <th>is bold by default.
9. **<td>**:this element defines a standard data cell in a table.
10. **rowspan**: The rowspan attribute is used to merge a cell vertically across multiple rows.
11. **colspan**: The colspan attribute is used to merge a cell horizontally across multiple columns
12. ****:this element is used to make text bold

WEEK-4

AIM:

A college wants to collect student details through an online admission form.design a registration form using only HTML tags with proper labels ,table alignment ,background colour,and border

CODE:

```
<!DOCTYPE html>
<html>
<body style="background-color: lavender; color: black;text-align: left"></body>
  <h1 style="font-family: Verdana, Geneva, Tahoma, sans-serif;font-size: 100%; text-align: center;">College Admission Registration</h1>
```

```

<table width="600" border="1"
style="background-color: white; height: 500px;margin: auto;">
<tr>
  <th>Full-name:</th>
  <th><input type="text"></th>
</tr>
<tr>
  <th>E-mail:</th>
  <th><input type="text"></th>
</tr>
<tr>
  <th>PHONE_NUMBER:</th>
  <th><input type="text"></th>
</tr>
<tr>
  <th>GENDER:</th>
  <th>
    <input type="radio" id="female" name="gender" value="female" >
    <label for="female">Female</label>
    <input type="radio" id="male" name="gender" value="male">
    <label for="male">Male</label>
    <input type="radio" id="others" name="gender" value="others">
    <label for="others">Others</label>
  </th>
</tr>
<tr>
  <th>COURSE APPLYING FOR:</th>
  <th>
    <!--<select id="course" name="course" size="1">-->
    <select id="course" name="course">
    <option value="">--select--</option>

```

```

        <option value="B-Tech">B-Tech</option></th>
    </tr>
    <tr>
        <th>HOBBIES:</th>
        <th>
            <input type="checkbox" id="sports" name="hobbies" value="sports" >
            <label for="sports">sports</label>
            <input type="checkbox" id="music" name="hobbies" value="music">
            <label for="music">music</label>
            <input type="checkbox" id="reading" name="hobbies" value="reading">
            <label for="reading">reading</label>
        </th>
    </tr>
    <tr>
        <th>DATE OF BIRTH:</th>
        <th><input type="date"></th>
    </tr>
    <tr>
        <th>UPLOAD CERTIFICATE:</th>
        <th><input type="file"></th>
    </tr>
    <tr>
        <th>ADRESS:</th>
        <th><textarea name="message" rows="8" cols="20"></textarea></th>
    </tr>
    <tr>
        <th colspan="2" style="text-align: center;">
            <input type="submit">
            <input type="reset">
        </th>
    </tr>
</tr>

```

</html>

Explanation of tags:

<form>:tag is used to create a form to collect user information.

2. <label>: The tag defines a label for an input field.

3. <input>: The tag is used to create input fields in a form. It is an empty tag. Types of Input Used:

4. type="radio": Used to select one option from multiple choices.

5.type="checkbox": Used to select multiple options.

6. type="file": Used to upload files.

7. type="submit": Used to submit the form.

8. type="reset": Used to clear the form data.

9. size: The size attribute specifies the width of the input field.

10. <textarea>: The tag is used to create a multi-line text input field.

College Admission Registration

Full-name:	
E-mail:	
PHONE_NUMBER:	
GENDER:	☐ Female ☐ Male ☐ Others
COURSE APPLYING FOR:	--select--
HOBBIES:	☐ sports ☐ music ☐ reading
DATE OF BIRTH:	dd-mm-yyyy
UPLOAD CERTIFICATE:	Choose File No file chosen
ADDRESS:	
Submit Reset	

B).AIM:

A company wants to design a webpage that displays two different forms side by side using iframes. One Iframes should display a Login Form and the other should display a Registration Forms.

CODE:

```
<!DOCTYPE html>
```

```
<html>
```

```
<body style="background-color:peachpuff; color: black;text-align: left">
```

```
<h1 style="font-family: Verdana, Geneva, Tahoma, sans-serif;font: size 100%; text-align: center;">TWO FORMS DISPLAYED USING IFRAMES</h1>
```

```
</body>
```

```
<table align="center" cellpadding="20">
```

```
<tr>
```

```
<!-- First Table -->
```

```
<td border: 2px solid black;>
```

```
<table border="1" cellpadding="10">
```

```
<tr>
```

```
<th colspan="2">User Login</th>
```

```
</tr>
```

```
<tr>
```

```
<td>Username:</td>
```

```
<td><input type="text"></td>
```

```
</tr>
```

```
<tr>
```

```
<td>Password:</td>
```

```
<td><input type="password"></td>
```

```
</tr>
```

```
<tr>
```

```
<td colspan="2" align="center">
```

```
<input type="button" value="Login">
```

```
</td>
```

```
</tr>
```

```
</table>
```

```
</td>
```

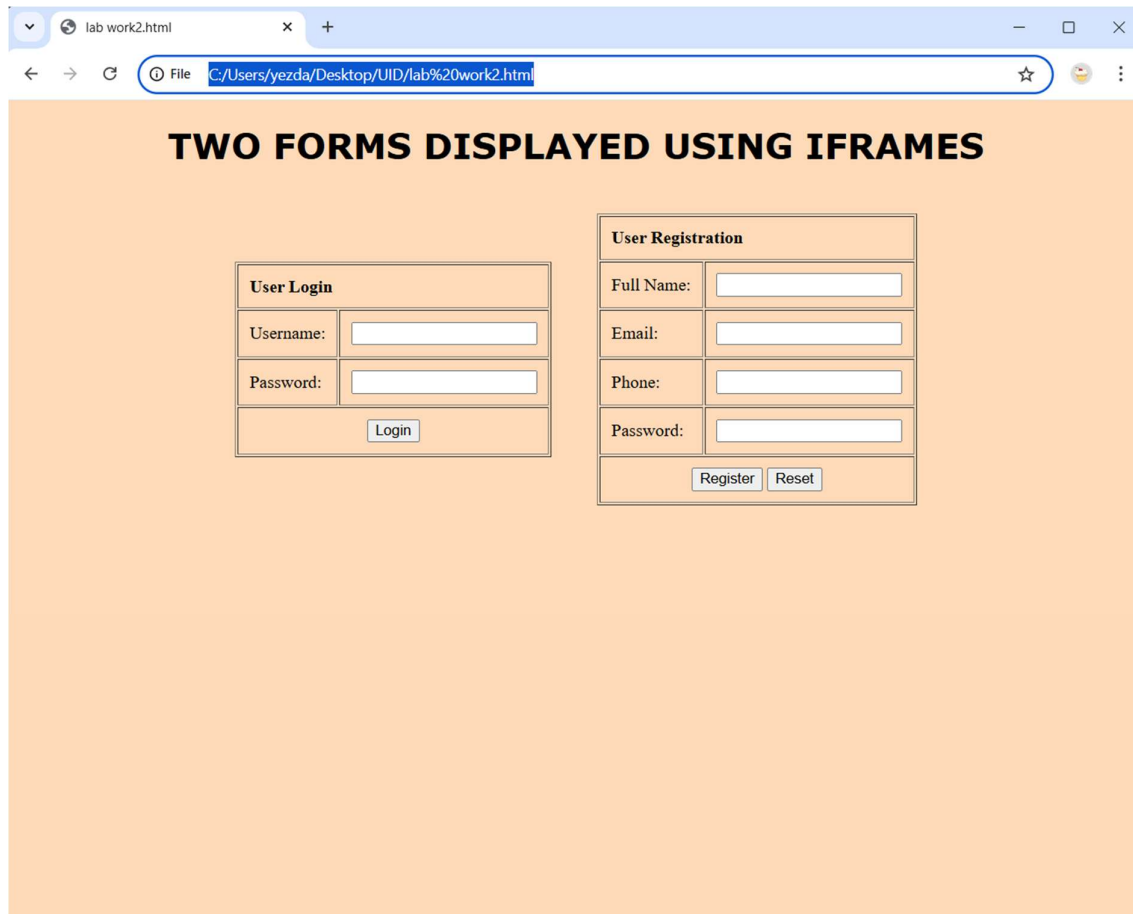
```
<!-- Second Table -->
```

```
<td border: 2px solid black;>
```

```
<table border="1" cellpadding="10">
```

```
<tr>
```

```
        <th colspan="2">User Registration</th>
    </tr>
    <tr>
        <td>Full Name:</td>
        <td><input type="text"></td>
    </tr>
    <tr>
        <td>Email:</td>
        <td><input type="text"></td>
    </tr>
    <tr>
        <td>Phone:</td>
        <td><input type="text"></td>
    </tr>
    <tr>
        <td>Password:</td>
        <td><input type="password"></td>
    </tr>
    <tr>
        <td colspan="2" align="center">
            <input type="button" value="Register">
            <input type="reset" value="Reset">
        </td>
    </tr>
</table>
</td>
</tr>
</table>
```



Explanation of tags:

**
**:the
 tag is used to insert a line break it is an empty tag.

Align:the align attribute is used to align content .

<iframe>:the <iframe>tag is used to display another webpage or content inside the current webpage.

Srdoc:the srdoc attribute is used display HTML content directly inside the iframe

WEEK-5

1.Aim : Create a card layout using tag. Consider three cards and a button for each:

UID25242-SK.TASMIYA

<!DOCTYPE html>

<html>

<head>

```
<title>Card Layout</title>
```

```
<style>
```

```
body{
```

```
font-family: Arial;
```

```
text-align: center;
```

```
}
```

```
.container{
```

```
display: flex;
```

```
flex-wrap: wrap;
```

```
justify-content: center;
```

```
gap: 20px;
```

```
padding: 20px;
```

```
}
```

```
.card{
```

```
width: 250px;
```

```
border: 2px solid black;
```

```
background-color: white;
```

```
padding: 15px;
```

```
}
```

```
.card img{
```

```
width: 100%;
```

```
height: 150px;
```

```
}
```

```
button{
```

```
padding: 10px 20px;
```

```
background-color: blue;
```

```
color: white;
```

```
border: none;
```

```
cursor: pointer;
```

```
}
```

```
button:hover{
```

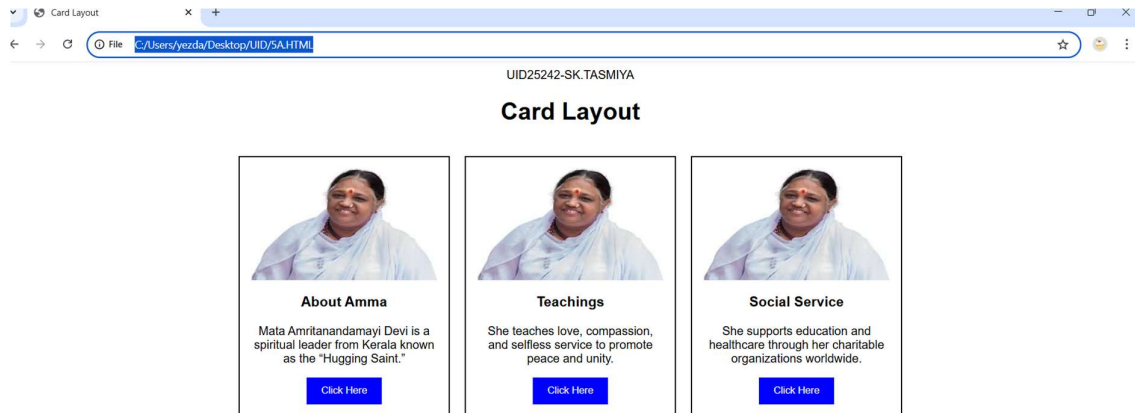
```
color:rgb(240, 240, 240);
}
</style>
</head>
<body>
<h1>Card Layout</h1>
<div class="container">
<div class="card">

<h3>About Amma</h3>
<p>Mata Amritanandamayi Devi is a spiritual leader from Kerala known as the “Hugging Saint.”</p>
<button>Click Here</button>
</div>
<div class="card">

<h3>Teachings</h3>
<p>She teaches love, compassion, and selfless service to promote peace and unity.</p>
<button>Click Here</button>
</div>
<div class="card">

<h3>Social Service</h3>
<p>She supports education and healthcare through her charitable organizations worldwide.</p>
<button>Click Here</button>
</div>
</div>
</body>
</html>
```

OUTPUT:



Explanation of the tags :

- 1.<body> : Contains all visible content of the webpage.
- 2.<h1> : Defines the main heading.
- 3.<style> : Used to write internal CSS for styling.
- 4. : Displays an image on the webpage.
- 5.<div> : Container element used to group content.
- 6.src : – Attribute of ,that specifies the image file path or URL.
- 7.<h3> : Defines a smaller heading inside each card.
- 8.<p> : Defines a paragraph of text.
- 9.<button>: Creates a clickable button.

2.Aim : Create a mini website using iframe with navigation (Home, About, Contact) that display content in the same page.

UID25242-SK.TASMIYA

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Mini Website</title>
```

```
<style>
*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:Arial, sans-serif;
}
body{
background: linear-gradient(to right, #4e73df, #1cc88a);
text-align:center;
color:white;
}
header{
padding:20px;
font-size:30px;
font-weight:bold;
}
nav{
background: rgba(0,0,0,0.3);
padding:15px;
}
nav a{
color:white;
text-decoration:none;
margin:0 25px;
font-size:18px;
padding:8px 15px;
border-radius:20px;
transition:0.3s;
}
nav a:hover{
background:white;
color:#4e73df;
}
iframe{
width:80%;
height:400px;
margin-top:30px;
border:none;
border-radius:15px;
background:white;
box-shadow:0 10px 20px rgba(0,0,0,0.3);
}
</style>
</head>
<body>
<header>Mini Website</header>
```



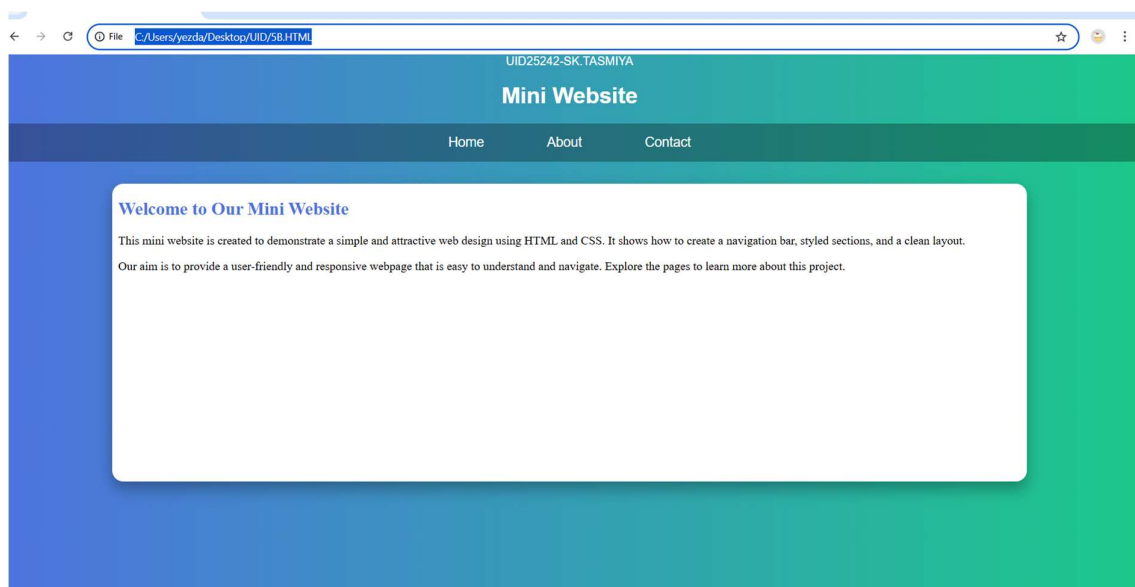
```

<nav>
<a href="#home"
onclick="document.getElementById('frame').srcdoc=document.getElementById('home').innerHTMLHTML;return false;">Home</a>
<a href="#about"
onclick="document.getElementById('frame').srcdoc=document.getElementById('about').innerHTMLHTML;return false;">About</a>
<a href="#contact"
onclick="document.getElementById('frame').srcdoc=document.getElementById('contact').innerHTMLHTML;return false;">Contact</a>
</nav>
<iframe id="frame"></iframe>
<div id="home" style="display:none;">
<h2 style="color:#4e73df;">Welcome to Our Mini Website</h2>
<p>This mini website is created to demonstrate a simple and attractive web design using HTML and CSS. It shows how to create a navigation bar, styled sections, and a clean layout.</p>
<p>Our aim is to provide a user-friendly and responsive webpage that is easy to understand and navigate. Explore the pages to learn more about this project.</p>
</div>
<div id="about" style="display:none;">
<h2 style="color:#1cc88a;">About This Website</h2>
<p>This website is a basic web development project designed for learning purposes. It explains how to structure a webpage using HTML and apply styling using CSS.</p>
<p>The project includes:</p>
<ul>
<li>Navigation Menu</li>
<li>Styled Content Sections</li>
<li>Responsive Layout</li>
<li>Clean and Modern Design</li>
</ul>
<p>This mini project helps beginners understand the fundamentals of web design.</p>
</div>
<div id="contact" style="display:none;">
<h2 style="color:#e74a3b;">Contact Us</h2>
<p>If you have any questions or suggestions regarding this website, feel free to contact us.</p>
<p>Email: example@email.com</p>
<p>Phone: +91 98765 43210</p>
<p>Location: Andhra Pradesh, India</p>
<p>We appreciate your feedback and support. Thank you for visiting our mini website!</p>

```

```
</div>
<script>
document.getElementById('frame').srcdoc=document.getElementById('home').innerHTML;
</script>
</body>
</html>
```

OUTPUT:



Explanation of the tags :

*(universal selector) : Applies CSS styles to all elements.

<nav>: Defines the navigation menu section.

<a> : Anchor tag used to create clickable links.

<onclick> : Attribute that runs JavaScript code when the link is clicked.

href : Attribute that specifies the link target (here used with section IDs).

Return false : Prevents the default link action (page jump).

Srcdoc : Attribute of used to insert HTML content directly into it.

<iframe> : – Embeds another HTML content inside the same webpage.

<div> : Container element used to group hidden content sections.

Id : Attribute used to uniquely identify an element.

Style : "display:none;" – Inline CSS used to hide content sections.

 : Creates an unordered (bulleted) list.

 : Defines individual list items inside a list.

3. Aim :

Design an amrita login page showing different campuses on the homepage. On clicking each campus, display campus details on the same page using HTML and CSS to design stylish webpage.

UID25242-SK.TASMIYA

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Amrita Vishwa Vidyapeetham</title>
```

```
<style>
```

```
*{
```

```
margin:0;
```

```
padding:0;
```

```
box-sizing:border-box;
```

```
font-family:'Times New Roman', serif;
```

```
}
```

```
body{
```

```
margin:0;
```

```
min-height:100vh;
```

```
background-image:url('Amravati-02.jpg');
```

```
background-size:cover;
```

```
background-position:center;
```

```
background-repeat:no-repeat;
```

```
background-attachment:fixed;
```

```
}
```

```
.header{
background:#8b0000;
color:white;
text-align:center;
font-size:42px;
padding:20px;
letter-spacing:3px;
}
input[type="radio"]{
display:none;
}
.menu{
background:#1e90ff;
text-align:center;
padding:10px;
}
.menu label{
color:yellow;
font-size:17px;
margin:8px 12px;
cursor:pointer;
font-weight:bold;
}
.menu label:hover{
color:white;
}
.container{
display:flex;
justify-content:space-around;
padding:40px;
flex-wrap:wrap;
```

```
}  
.login-box{  
width:320px;  
padding:30px;  
border-radius:15px;  
background:linear-gradient(to bottom,#1e90ff,#32cd32);  
color:white;  
box-shadow:0 10px 20px rgba(0,0,0,0.6);  
}  
.login-box h2{  
text-align:center;  
margin-bottom:20px;  
text-decoration:underline;  
}  
.login-box input{  
width:100%;  
padding:8px;  
margin:10px 0;  
border-radius:5px;  
border:none;  
}  
.remember{  
display:flex;  
justify-content:space-between;  
align-items:center;  
margin:10px 0;  
font-size:14px;  
}  
.save{  
display:flex;  
align-items:center;
```

```
gap:5px;
white-space:nowrap;
}
.login-box a{
color:blue;
}
.content{
width:650px;
background:white;
padding:25px;
border-radius:10px;
box-shadow:0 10px 20px rgba(0,0,0,0.6);
display:none;
}
.content h2{
color:#8b0000;
text-align:center;
margin-bottom:15px;
}
.content img{
width:100%;
height:280px;
margin-top:15px;
border-radius:10px;
}
#amaravati:checked ~ .container #amaravatiContent,
#amritapuri:checked ~ .container #amritapuriContent,
#bengaluru:checked ~ .container #bengaluruContent,
#chennai:checked ~ .container #chennaiContent,
#coimbatore:checked ~ .container #coimbatoreContent,
#faridabad:checked ~ .container #faridabadContent,
```

```
#kochi:checked ~ .container #kochiContent,
#mysuru:checked ~ .container #mysuruContent{
display:block;
}
</style>
</head>
<body>
<div class="header">
AMRITA VISHWA VIDYAPEETHAM
</div>
<input type="radio" name="campus" id="amaravati">
<input type="radio" name="campus" id="amritapuri">
<input type="radio" name="campus" id="bengaluru">
<input type="radio" name="campus" id="chennai">
<input type="radio" name="campus" id="coimbatore">
<input type="radio" name="campus" id="faridabad">
<input type="radio" name="campus" id="kochi">
<input type="radio" name="campus" id="mysuru">
<div class="menu">
<label for="amaravati">AMARAVATI</label>
<label for="amritapuri">AMRITAPURI</label>
<label for="bengaluru">BENGALURU</label>
<label for="chennai">CHENNAI</label>
<label for="coimbatore">COIMBATORE</label>
<label for="faridabad">FARIDABAD</label>
<label for="kochi">KOCHI</label>
<label for="mysuru">MYSURU</label>
</div>
<div class="container">
<div class="login-box">
<h2>Login</h2>
```

```
<input type="email" placeholder="Enter your email id">
<input type="password" placeholder="Enter your password">
<div class="remember">
<label class="save">
<input type="checkbox"> Save me
</label>
<a href="#">Forgot Password?</a>
</div>
<button>Login</button>
<p>Don't have an account? <a href="#">Register</a></p>
</div>
<div class="content" id="amaravatiContent">
<h2>About Amaravati Campus</h2>
<p>Amrita Vishwa Vidyapeetham Amaravati Campus is located in Andhra Pradesh.
It offers quality education with modern infrastructure and advanced laboratories.
The campus provides a peaceful environment that supports learning and research.
It focuses on academic excellence, innovation, and overall student development.</p>

</div>
<div class="content" id="amritapuriContent">
<h2>About Amritapuri Campus</h2>
<p>Amrita Vishwa Vidyapeetham Amritapuri Campus is located in Kerala near the
Arabian
Sea.
It is the headquarters of the university and offers a peaceful learning environment.
The campus provides courses in engineering, arts, science, and social work.
It focuses on academic excellence along with spiritual and value-based education.</p>

</div>
<div class="content" id="bengaluruContent">
<h2>About Bengaluru Campus</h2>
```


<p>The Bengaluru Campus is known for its strong focus on engineering and technology programs.

It offers modern laboratories and research facilities.

The campus encourages innovation and industry collaboration.

It provides students with excellent academic and placement opportunities.</p>

</div>

<div class="content" id="chennaiContent">

<h2>About Chennai Campus</h2>

<p>The Chennai Campus specializes mainly in engineering education.

It provides advanced infrastructure and technical training.

The campus promotes research and practical learning.

It prepares students for successful careers in technology fields.</p>

</div>

<div class="content" id="coimbatoreContent">

<h2>About Coimbatore Campus</h2>

<p>The Coimbatore Campus is one of the largest campuses of the university.

It offers diverse programs in engineering, management, arts, and science.

The campus has excellent research centers and modern facilities.

It focuses on academic excellence and holistic student development.</p>

</div>

<div class="content" id="faridabadContent">

<h2>About Faridabad Campus</h2>

<p>The Faridabad Campus is located near New Delhi.

It offers programs in engineering, management, and medical sciences.

The campus emphasizes research and innovation.

It provides strong industry exposure and career support.</p>

</div>

<div class="content" id="kochiContent">

<h2>About Kochi Campus</h2>

<p>The Kochi Campus is well known for medical and health science programs.

It includes a teaching hospital with advanced healthcare facilities.

The campus provides quality education in medicine and nursing.

It focuses on service-oriented and value-based learning.</p>

</div>

<div class="content" id="mysuruContent">

<h2>About Mysuru Campus</h2>

<p>The Mysuru Campus offers programs in engineering and postgraduate studies.

It provides a calm and focused learning environment.

The campus supports research and skill development.

It encourages students to achieve academic and professional excellence.</p>

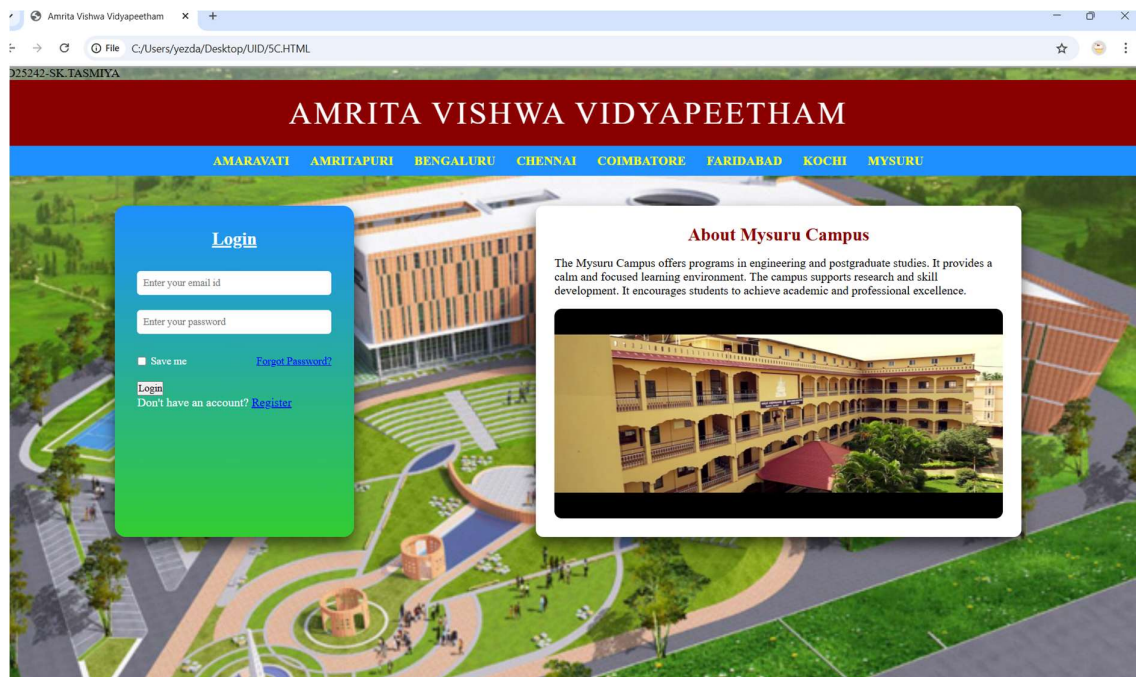
</div>

</div>

</body>

</html>

OUTPUT



Explanation of the tags :-

<input>: Creates input fields such as radio buttons, email, password, and checkbox.

type="radio" – Creates radio buttons for selecting campuses. **type="email"** – Creates an email input field with validation. **type="password"** – Creates a password input field that hides typed text.

type="checkbox" – Creates a checkbox (used for “Save me”). **placeholder** – Displays hint text inside input fields.

<label>: Defines a clickable label for form controls (used for campus selection).

<button>: Creates a clickable Login button.

WEEK-6

AIM: DESIGN SIMPLE A WEBSITE OF E-COMMERCE STORE USING HTML

CODE:

```
<!DOCTYPE html>
```

```
<html>
<head>
<title>SIMPLE E-COMMERCE STORE</title>
<style>
.container{
  display:grid;
  grid-template-rows: auto auto 1fr auto;
  grid-template-columns:200px 1fr;
  grid-template-areas:
    "header header"
    "sale sale"
    "sidebar products"
    "footer footer";
}
.header{
  grid-area: header;
  text-align:center;
  padding:10px;
}
.sale{
  grid-area:sale;
  text-align:center;
  padding:10px;
}
.sidebar{
  grid-area:sidebar;
  padding:10px;
  border:1px solid black;
}

.sidebar a{
```

```
display:block;
text-decoration:none;
color:black;
padding:5px;
border-bottom:1px solid black;
}
.products{
  grid-area:products;
  padding:10px;
}
.section{
  display:none;
}
.section:target{
  display:grid;
  grid-template-columns:repeat(2,1fr);
  gap:15px;
}
.product{
  border:1px solid black;
  padding:10px;
  text-align:center;
}
.footer{
  grid-area:footer;
  background:black;
  color:white;
  text-align:center;
  padding:10px;
}
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<div class="header">
```

```
<h2>My E-Commerce Store</h2>
```

```
</div>
```

```
<div class="sale">
```

```
<h3>Big Sale upto 50% off</h3>
```

```
</div>
```

```
<div class="sidebar">
```

```
<h3>Categories</h3>
```

```
<a href="#electronics">Electronics</a>
```

```
<a href="#clothes">Clothes</a>
```

```
<a href="#home">Home Appliances</a>
```

```
<a href="#books">Books</a>
```

```
</div>
```

```
<div class="products">
```

```
<!-- Electronics -->
```

```
<div id="electronics" class="section">
```

```
<div class="product">
```

```
<p>handfan</p>
```

```

```

```
<p>$49</p>
```

```
<button>Add to Cart</button>
```

```
</div>
<div class="product">
  <p>headphones</p>
  
  <p>$79</p>
  <button>Add to Cart</button>
</div>
<div class="product">
  <p>instax mini 12</p>
  
  <p>$80</p>
  <button>Add to Cart</button>
</div>
<div class="product">
  <p>Smart Watch</p>
  
  <p>$89</p><button>Add to Cart</button>
</div>
</div>

<!-- Clothes -->
<div id="clothes" class="section">
  <div class="product">
    <p>T-Shirt</p>
    
    <p>$49</p><button>Add to Cart</button>
  </div>
  <div class="product">
    <p>Jeans</p>
    
    <p>$79</p>
```

```
<button>Add to Cart</button>
</div>
<div class="product">
<p>Jacket</p>

<p>$80</p>
<button>Add to Cart</button>
</div>
<div class="product">
<p>Shoes</p>

<p>$89</p>
<button>Add to Cart</button>
</div>
</div>
```

```
<!-- Home Appliances -->
<div id="home" class="section">
<div class="product">
<p>Fan</p>

<p>$49</p>
<button>Add to Cart</button>
</div>
<div class="product">
<p>TV</p>

<p>$79</p>
<button>Add to Cart</button>
</div>
<div class="product">
```



```
<p>Fridge</p>

<p>$80</p>
<button>Add to Cart</button>
</div>
<div class="product">
<p>Washing Machine</p>

<p>$89</p>
<button>Add to Cart</button>
</div>
</div>

<!-- Books -->
<div id="books" class="section">
<div class="product">
<p>Math Book</p>

<p>$49</p>
<button>Add to Cart</button>
</div>
<div class="product">
<p>Science Book</p>

<p>$79</p>
<button>Add to Cart</button>
</div>
<div class="product">
<p>Story Book</p>

<p>$80</p>
```

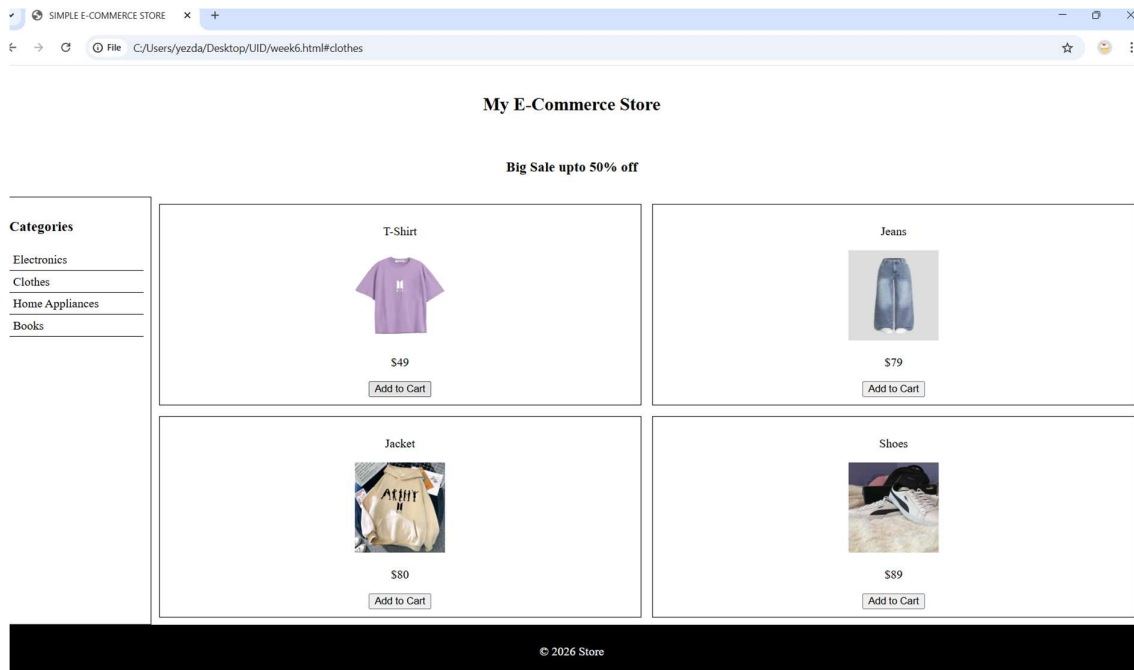
```
<button>Add to Cart</button>
</div>
<div class="product">
<p>History Book</p>

<p>$89</p>
<button>Add to Cart</button>
</div>
</div>

<div class="footer">
<p>© 2026 Store</p>
</div>
w
</div>

</body>
</html>
```

OUTPUT:



Explanation of tags:

- 1).<header>:The branding area at the top.
- 2) .sale- An announcement bar for marketing.
- 3) .categories- A sidebar acting as the "Navigation Menu."
- 4) .products-A grid container that holds individual item cards.
- 5) Footer-Copyright and legal info at the bottom.

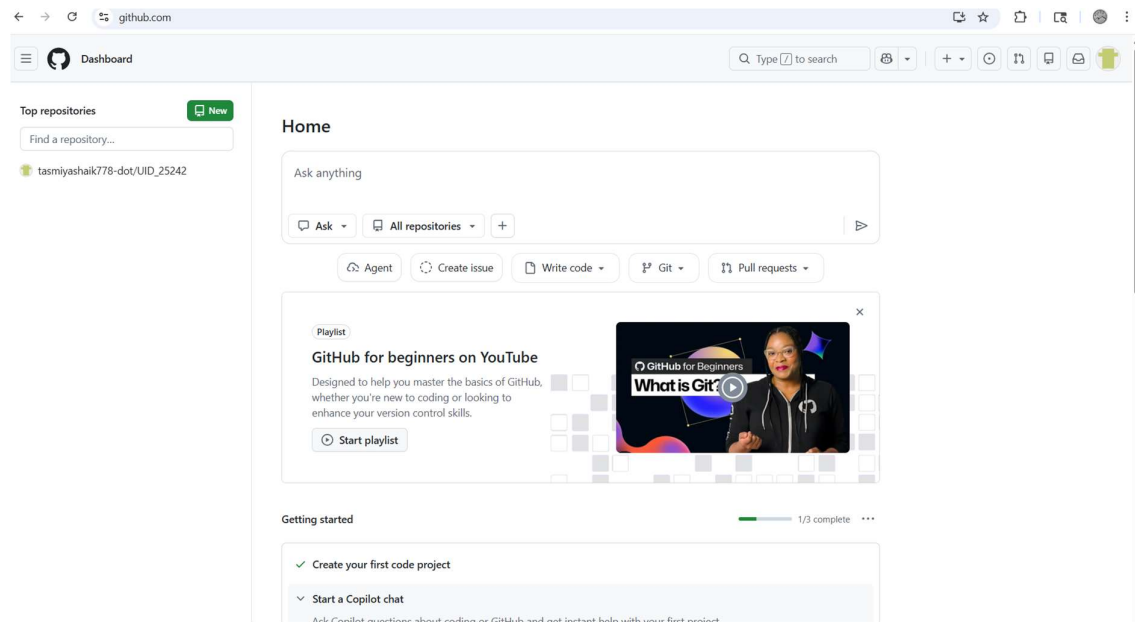
WEEK-7

AIM: To create a new repository in git

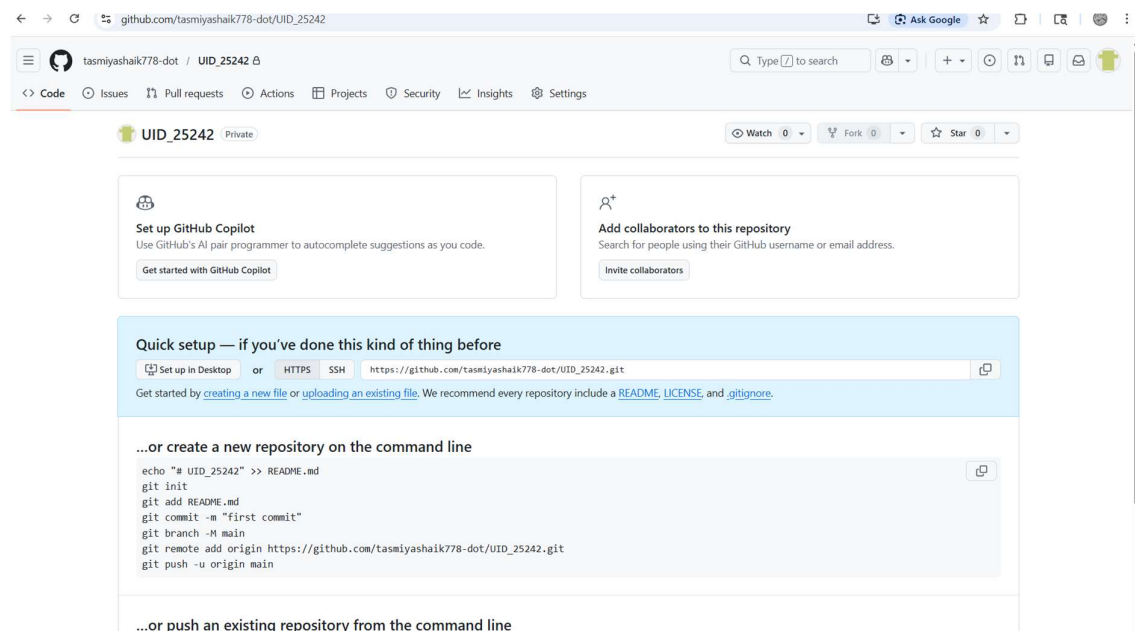
PROCEDURE:

Step 1 : Install the git and open your account

Step 2 :



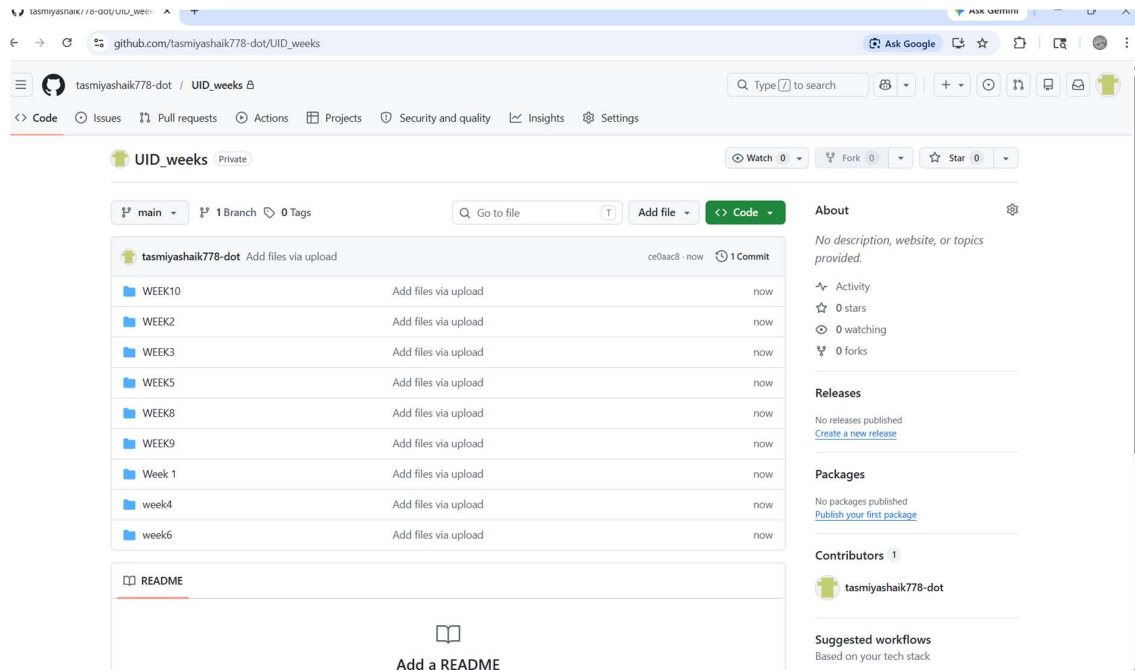
Step 3 : Click on the new repository



Step 4 : Now give a name to your repository and select an existing file or any file to upload in the repository

Step 5 :After adding files to your repository , copy the code to your git repository path for future reviewing .

Step 6 : After creating the repository you can go to my repositories and then cross check your repository to see which files you have added like this



So , now you have successfully created your repository and added your files to it .

WEEK-8:

AIM:

Create a webpage that display a square that continuously rotates 360 degrees using css.

CODE:

UIDCSE25242-SK.TASMIYA

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Rotating Square</title>
```

```
<style>
  body.square {
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
    margin: 0;
    background-color: #f0f0f0;
  }

  .square {
    width: 150px;
    height: 150px;
    background-color: Blue;
    animation: rotateSquare 2s linear infinite;
  }

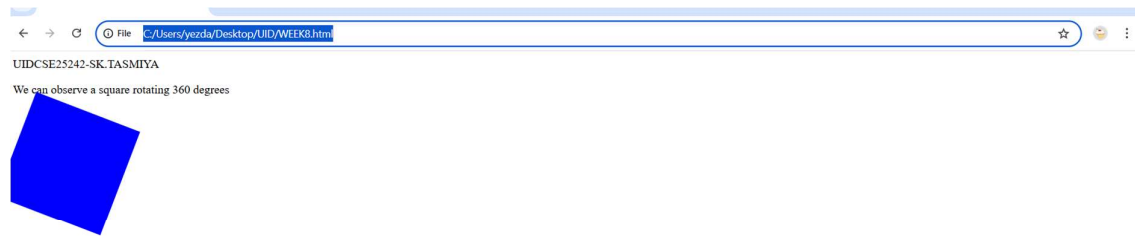
  @keyframes rotateSquare {
    from {
      transform: rotate(0deg);
    }
    to {
      transform: rotate(360deg);
    }
  }
</style>
</head>
<body>
  <p>We can observe a square rotating 360 degrees</p>

  <div class="square"></div>
```

```
</body>
```

```
</html>
```

OUTPUT:



Explanation of tags:

- 1) **@keyframes** as the storyboard or the script for an animation.
- 2) **background-attachment: fixed**: This is a nice touch that keeps the background image from moving if your page ends up having a scrollbar.
- 3) **display: flex**: This is the modern "basic" way to perfectly center an element (like our square) both horizontally and vertically.
- 4) **height: 100vh**: This tells the body to take up 100% of the viewport height so the centering works correctly.

b).

AIM: Create a ball that moves up and down continuously using css animation.

CODE:

```
UIDCSE25242-SK.TASMIYA
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Moving Ball</title>
```

```
<style>
body {
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
  margin: 0;
  background-color: wheat;
}

.ball {
  width: 80px;
  height: 80px;
  background-color: pink;
  border-radius: 50%;
  animation: bounce 1s ease-in-out infinite;
}

@keyframes bounce {
  0% {
    transform: translateY(0);
  }
  50% {
    transform: translateY(-150px);
  }
  100% {
    transform: translateY(0);
  }
}
</style>
</head>
```



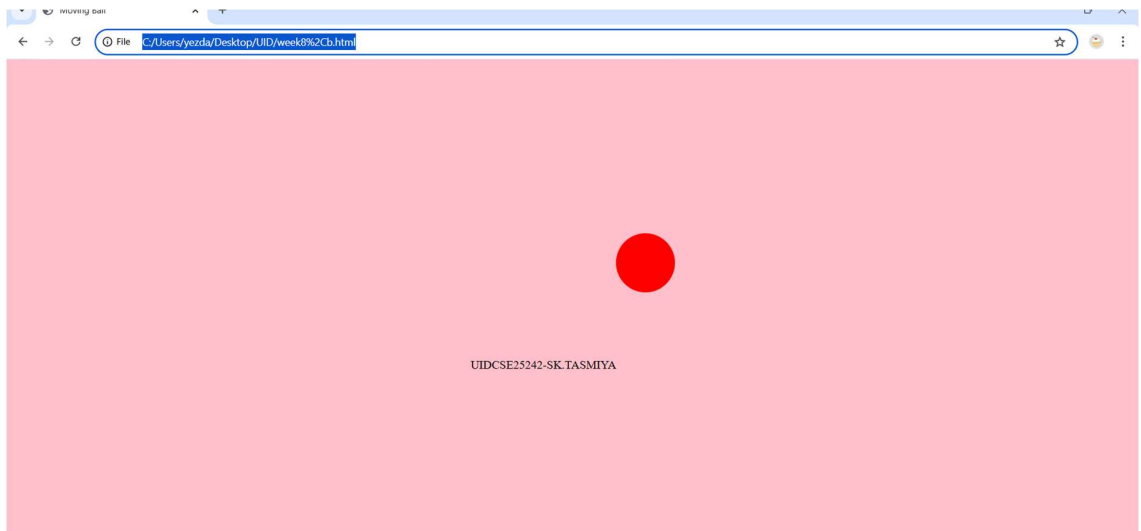
```
<body>
```

```
<div class="ball"></div>
```

```
</body>
```

```
</html>
```

OUTPUT:



Explanation of tags:

1). Instead of margin-top, use transform: translateY(). This allows the GPU to handle the animation, resulting in a much silkier 60fps movement.

2). Currently, the ball moves at a constant speed (linear). By adding an ease-in timing function, you can make it look like gravity is pulling it down, making the bounce feel "weighty."

3),. Inside the .ball class, we define what the object looks like before it starts moving:

- width & height: Sets the size.
- background-color: Gives it that punchy red color.
- border-radius: 50%: This is the magic line that turns a square div into a perfect circle.
- box-shadow: This is optional, but it adds a tiny bit of depth so the ball doesn't look flat against the background.

C).AIM: Create a circle that moves horizontally and rotates at the same time using CSS animation

CODE:

UIDCSE25242-SK.TASMIYA

```
<!DOCTYPE html>
<html>
<head>
<style>
.ball {
width: 50px;
height: 50px;
background-color: pink;
border-radius: 50%;
animation-name: moveUpDown;
animation-duration: 1s;
animation-iteration-count: infinite;
animation-direction: alternate; }
@keyframes moveUpDown {
from {
margin-top: 0px;
}
to {
margin-top: 200px; }
}
</style>
</head>
<body>
<div class="ball"></div>
</body>
</html>
```

OUTPUT:



Explanation of tags:

1. animation: This shorthand tells the browser: "Run the move animation for 3 seconds, keep it steady (linear), do it forever (infinite), and go back and forth (alternate)."
2. rotate: This is a modern CSS property. By keeping it separate from transform, we can animate the spinning and the sliding at the same time without the code getting messy.
3. border-top: Since a solid circle looks the same when it spins, the top border acts as a "marker" so you can actually see the rotation happening.

WEEK-9

AIM:

create an webpage usinig html,css,javascript,based on the given conditions:

- a)create a button for each program based on the button clicked the corresponding program has to be excuted and should display the output
- b)the logics for each button includes finding an even number, odd number, prime or not ,factorial of a number , armstrong number use appropriate designs to display the output.

CODE:

UIDCSE25242-SK.TASMIYA

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Number Operations</title>
```

```
  <style>
```

```
body {  
  font-family: Verdana, Geneva, Tahoma, sans-serif;  
  background-color: lightpink;  
  text-align: center;  
  padding: 50px;  
}
```

```
.container {  
  background: white;  
  padding: 50px;  
  border-radius: 10px;  
  width: 500px;  
  margin: auto;  
  box-shadow: 0px 0px 15px rgba(0,0,0,0.2);  
}
```

```
input {  
  padding: 10px;  
  width: 80%;  
  margin-bottom: 20px;  
  border-radius: 5px;  
  border: 1px solid gray;  
}
```

```
button {  
  margin: 5px;  
  padding: 10px 15px;  
  border: none;  
  border-radius: 5px;  
  background-color: aquamarine;  
  color: white;
```

```
        cursor: pointer;
    }

    button:hover {
        background-color: greenyellow;
    }

    #output {
        margin-top: 20px;
        font-weight: bold;
        color: black;
    }
</style>
</head>

<body>

<div class="container">
    <h2>Number Operations</h2>

    <input type="number" id="num" placeholder="Enter a number">

    <br>

    <button onclick="checkEven()">Even</button>
    <button onclick="checkOdd()">Odd</button>
    <button onclick="checkPrime()">Prime</button>
    <button onclick="factorial()">Factorial</button>
    <button onclick="checkArmstrong()">Armstrong</button>

    <div id="output"></div>
```

```
</div>
```

```
<script>
```

```
function getNumber() {  
    let n = document.getElementById("num").value;  
    if (n === "") {  
        alert("Please enter a number");  
        return null;  
    }  
    return parseInt(n);  
}
```

```
function checkEven() {  
    let n = getNumber();  
    if (n !== null) {  
        document.getElementById("output").innerText =  
            (n % 2 === 0) ? n + " is Even" : n + " is NOT Even";  
    }  
}
```

```
function checkOdd() {  
    let n = getNumber();  
    if (n !== null) {  
        document.getElementById("output").innerText =  
            (n % 2 !== 0) ? n + " is Odd" : n + " is NOT Odd";  
    }  
}
```

```
function checkPrime() {  
    let n = getNumber();
```

```
if (n !== null) {
  if (n <= 1) {
    document.getElementById("output").innerText = n + " is NOT Prime";
    return;
  }

  let isPrime = true;
  for (let i = 2; i <= Math.sqrt(n); i++) {
    if (n % i === 0) {
      isPrime = false;
      break;
    }
  }

  document.getElementById("output").innerText =
    isPrime ? n + " is Prime" : n + " is NOT Prime";
}

function factorial() {
  let n = getNumber();
  if (n !== null) {
    if (n < 0) {
      document.getElementById("output").innerText = "Factorial not defined for negative numbers";
      return;
    }

    let fact = 1;
    for (let i = 1; i <= n; i++) {
      fact *= i;
    }
  }
}
```

```
    }

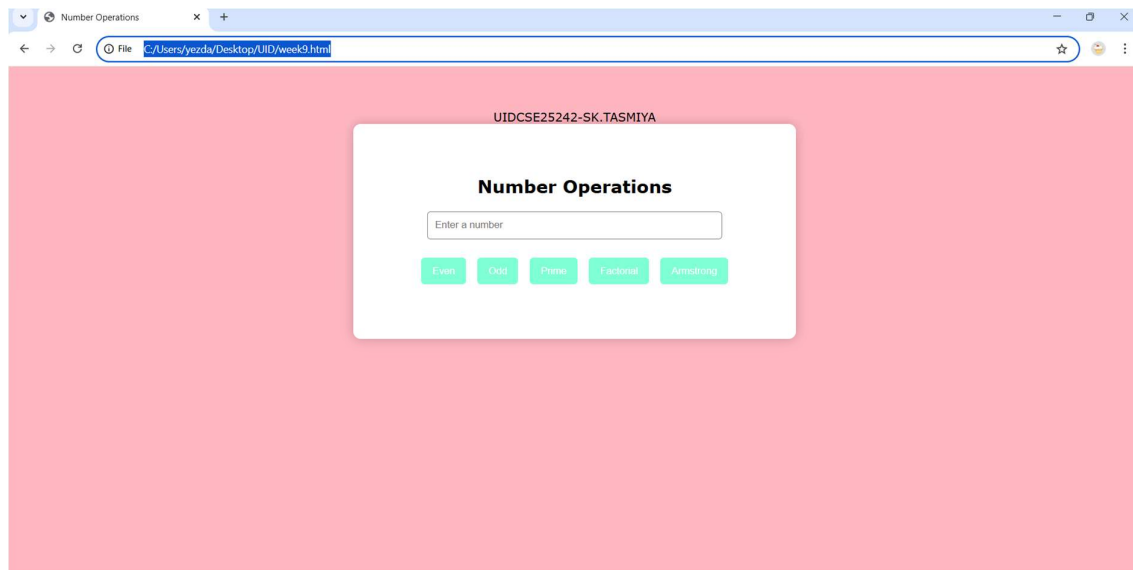
    document.getElementById("output").innerText =
        "Factorial of " + n + " is " + fact;
    }
}

function checkArmstrong() {
    let n = getNumber();
    if (n !== null) {
        let temp = n;
        let sum = 0;
        let digits = n.toString().length;

        while (temp > 0) {
            let rem = temp % 10;
            sum += Math.pow(rem, digits);
            temp = Math.floor(temp / 10);
        }

        document.getElementById("output").innerText =
            (sum === n) ? n + " is an Armstrong Number" : n + " is NOT an Armstrong Number";
    }
}
</script>
</body>
</html>
```

OUTPUT:



Explanation of tags:

<div>: Used to group elements together. In this program, it is used to create a container box for layout.

class="container": Assigns a class name to the div so that CSS styling can be applied.

<input type="number">: Creates an input box that allows the user to enter only numeric values.

id="num": Assigns an ID to the input field so it can be accessed using JavaScript.

<button>: Creates clickable buttons. Each button is used to perform a specific operation.

onclick: JavaScript event attribute. It calls a function when the button is clicked.

<div id="output">: Used to display the result/output of the selected operation.

border-radius: Rounds the corners of the container box.

box-shadow: Adds a shadow effect around the container for better design.

text-align: Aligns the text (e.g., center alignment).

function: A block of JavaScript code used to perform a specific task.

getElementById(): JavaScript method used to access HTML elements using their ID.

value: Retrieves the value entered by the user in the input field.

innerText: Used to display output inside an HTML element.

parseInt(): Converts the input value into an integer number.

if-else: Conditional statements used to check conditions and execute code accordingly.

% (modulus operator): Used to find remainder (helps in checking even/odd numbers).

for loop: Used to repeat a block of code multiple times (used in factorial and prime logic).

Math.sqrt(): Returns the square root of a number (used in prime number checking).

Math.pow(): Used to calculate power (used in Armstrong number logic).

WEEK-10

AIM:

A).write a java script conditional statement to find the sign of a product of three numbers using functions,display an alert box with specified sign.

CODE:

UIDCSE25242-SK.TASMIYA

```
<!DOCTYPE html>

<html>
<head>
  <title>Sign of Product</title>
</head>
<body>

<h2>Find Sign of Product</h2>

<label>Enter first number:</label>
<input type="number" id="num1"><br><br>

<label>Enter second number:</label>
<input type="number" id="num2"><br><br>

<label>Enter third number:</label>
```

```
<input type="number" id="num3"><br><br>

<button onclick="findSign()">Check Sign</button>

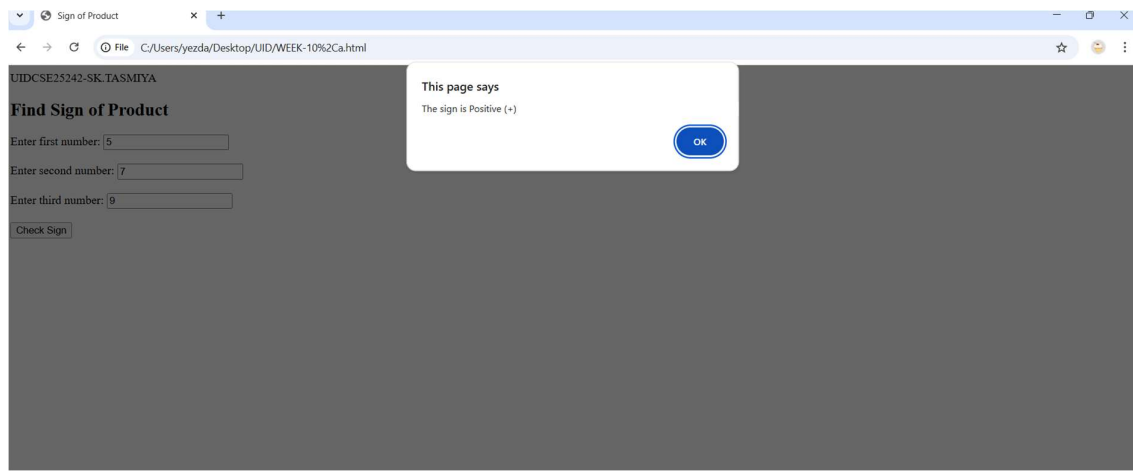
<script>

function findSign() {
    let a = parseInt(document.getElementById("num1").value);
    let b = parseInt(document.getElementById("num2").value);
    let c = parseInt(document.getElementById("num3").value);

    let product = a * b * c;

    if (product > 0) {
        alert("The sign is Positive (+)");
    }
    else if (product < 0) {
        alert("The sign is Negative (-)");
    }
    else {
        alert("The product is Zero");
    }
}
</script>
</body>
</html>
```

OUTPUT:



Explanation of tags:

<button>: Creates a clickable button.

onclick: Event attribute used to call a JavaScript function when the button is clicked.

<script>: Used to write JavaScript code inside the HTML document.

function: A block of code used to perform a specific task.

getElementById(): JavaScript method used to access HTML elements using their ID.

value: Retrieves the value entered in the input field.

parseInt(): Converts the input value into an integer.

if-else: Conditional statements used to check whether the product is positive, negative, or zero.

alert(): Displays the result in a popup message box.

B).AIM:

write a javascript conditional statement to sort three numbers using functions, display an alert box to show the result.

CODE:

UIDCSE25242-SK.TASMIYA

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Sort Three Numbers</title>
```

```
</head>
```

```
<body>
```

```
<h2>Sort Three Numbers</h2>
```

```
<label>Enter first number:</label>
```

```
<input type="number" id="num1"><br><br>
```

```
<label>Enter second number:</label>
```

```
<input type="number" id="num2"><br><br>
```

```
<label>Enter third number:</label>
```

```
<input type="number" id="num3"><br><br>
```

```
<button onclick="sortNumbers()">Sort Numbers</button>
```

```
<script>
```

```
function sortNumbers() {
```

```
    let a = parseInt(document.getElementById("num1").value);
```

```
    let b = parseInt(document.getElementById("num2").value);
```

```
    let c = parseInt(document.getElementById("num3").value);
```

```
    let result;
```

```
    if (a <= b && a <= c) {
```

```
        if (b <= c)
```

```
            result = a + ", " + b + ", " + c;
```

```
        else
```

```
            result = a + ", " + c + ", " + b;
```

```
    }
```

```
    else if (b <= a && b <= c) {
```

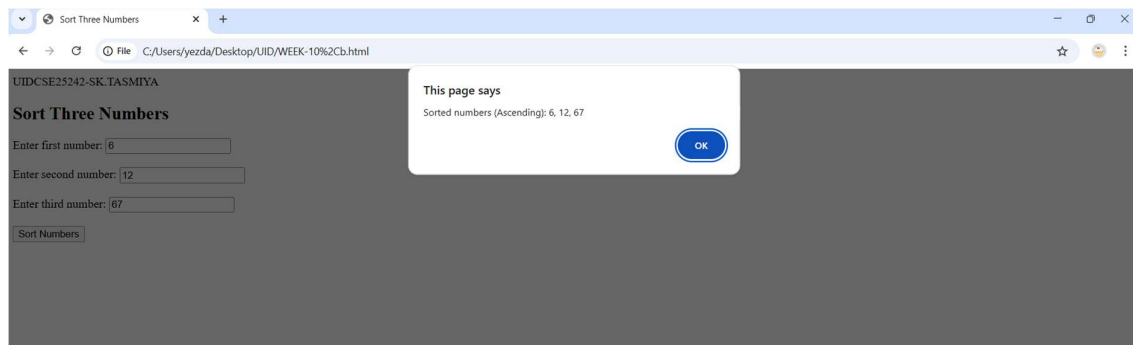
```
        if (a <= c)
```

```

        result = b + ", " + a + ", " + c;
    else
        result = b + ", " + c + ", " + a;
    }
    else {
        if (a <= b)
            result = c + ", " + a + ", " + b;
        else
            result = c + ", " + b + ", " + a;
        }
    alert("Sorted numbers (Ascending): " + result);
}
</script>
</body>
</html>

```

OUTPUT:



Explanation of tags:

<button>: Creates a clickable button to perform the sorting operation.

onclick: Event attribute that calls the JavaScript function when the button is clicked.

<script>: Contains JavaScript code within the HTML document.

function: Defines a block of code to perform a specific task (sorting numbers).

getElementById(): JavaScript method used to access HTML elements using their ID.

value: Retrieves the value entered in the input fields.

parseInt(): Converts the input values into integers.

if-else: Conditional statements used to compare numbers and arrange them in ascending order.

alert(): Displays the sorted result in a popup message box.

C).

AIM: write a javascript conditional statement to find the largest of five numbers using functions and onclick event, display an alert box to show the result

CODE:

UIDCSE25242-SK.TASMIYA

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Largest of Five Numbers</title>
```

```
</head>
```

```
<body>
```

```
<h2>Find Largest of Five Numbers</h2>
```

```
<label>Enter first number:</label>
```

```
<input type="number" id="num1"><br><br>
```

```
<label>Enter second number:</label>
```

```
<input type="number" id="num2"><br><br>
```

```
<label>Enter third number:</label>
```

```
<input type="number" id="num3"><br><br>
```

```
<label>Enter fourth number:</label>
```

```
<input type="number" id="num4"><br><br>
```

```
<label>Enter fifth number:</label>
```

```
<input type="number" id="num5"><br><br>
```

```
<button onclick="findLargest()">Find Largest</button>
```

```
<script>
```

```
function findLargest() {
```

```
    let a = parseInt(document.getElementById("num1").value);
```

```
    let b = parseInt(document.getElementById("num2").value);
```

```
    let c = parseInt(document.getElementById("num3").value);
```

```
    let d = parseInt(document.getElementById("num4").value);
```

```
    let e = parseInt(document.getElementById("num5").value);
```

```
    let largest = a;
```

```
    if (b > largest) {
```

```
        largest = b;
```

```
    }
```

```
    if (c > largest) {
```

```
        largest = c;
```

```
    }
```

```
    if (d > largest) {
```

```
        largest = d;
```

```
    }
```

```
    if (e > largest) {
```

```
        largest = e;
```

```
    }
```

```
    alert("Largest number is: " + largest);
```

```
}
```

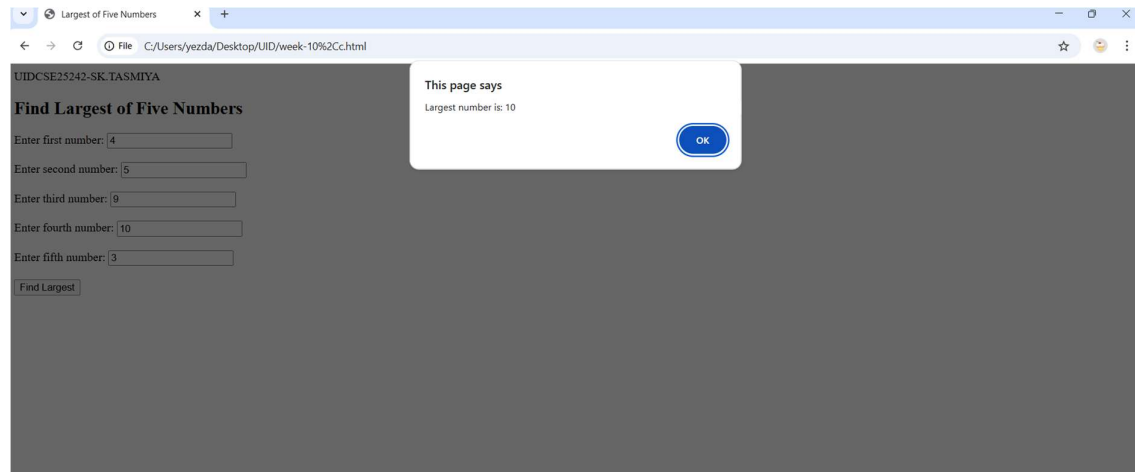


```
</script>
```

```
</body>
```

```
</html>
```

OUTPUT:



Explanation of tags:

<button>: Creates a clickable button.

onclick: Event attribute used to call the JavaScript function when the button is clicked.

<script>: Used to include JavaScript code inside the HTML file.

function: Defines a block of code that performs a specific task (finding the largest number).

getElementById(): JavaScript method used to access HTML elements using their ID.

value: Retrieves the value entered in the input fields.

parseInt(): Converts the input values into integers.

if: Conditional statement used to compare numbers and update the largest value.

alert(): Displays the final result in a popup message box.

D).

AIM: write a javascript for loop that will iterate from 0 to 15. for each iteration, it will check if the current number is even or odd and display a message to the screen.

CODE:

```
UIDCSE25242-SK.TASMIYA
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
  <title>Even or Odd (0 to 15)</title>
</head>
<body>
<h2>Check Even or Odd from 0 to 15</h2>
<button onclick="checkNumbers()">Show Result</button>
<div id="output"></div>
<script>
function checkNumbers() {
  let result = "";
  for (let i = 0; i <= 15; i++) {
    if (i % 2 === 0) {
      result += i + " is Even <br>";
    } else {
      result += i + " is Odd <br>";
    }
  }
}

document.getElementById("output").innerHTML = result;
}
</script>
</body>
</html>
```

OUTPUT:



Explanation of tags:

onclick: Event attribute used to call the JavaScript function when the button is clicked.

<div id="output">: Used to display the result/output on the screen.

<script>: Used to include JavaScript code inside the HTML file.

function: Defines a block of code that performs a specific task (checking even or odd numbers).

for loop: Used to iterate from 0 to 15 repeatedly.

if-else: Conditional statement used to check whether a number is even or odd.

% (modulus operator): Used to find the remainder when divided by 2 (helps in checking even/odd).

innerHTML: Used to display the output inside the <div> element.

getElementById(): JavaScript method used to access the output element using its ID.

E).

AIM: write a javascripts program to read 6 subjects marks from the student then compute the average marks of the student .then this average is used to marks the corresponding grades.

RANGE	GRADE
<60	F
<70	D
<80	C
<90	B
<100	A

CODE:

UIDCSE25242-SK.TASMIYA

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Grade</title>
</head>
<body>

<h2>Student Marks and Grade</h2>

<label>Subject 1:</label>
<input type="number" id="s1"><br><br>

<label>Subject 2:</label>
<input type="number" id="s2"><br><br>

<label>Subject 3:</label>
<input type="number" id="s3"><br><br>

<label>Subject 4:</label>
<input type="number" id="s4"><br><br>

<label>Subject 5:</label>
<input type="number" id="s5"><br><br>

<label>Subject 6:</label>
<input type="number" id="s6"><br><br>
<button onclick="calculateGrade()">Calculate Grade</button>
<div id="output"></div>
<script>
function calculateGrade() {
  let s1 = parseInt(document.getElementById("s1").value);
```

```
let s2 = parseInt(document.getElementById("s2").value);
let s3 = parseInt(document.getElementById("s3").value);
let s4 = parseInt(document.getElementById("s4").value);
let s5 = parseInt(document.getElementById("s5").value);
let s6 = parseInt(document.getElementById("s6").value);

let total = s1 + s2 + s3 + s4 + s5 + s6;
let avg = total / 6;
let grade;
if (avg >= 90)
    grade = "A+";
else if (avg >= 75)
    grade = "A";
else if (avg >= 60)
    grade = "B";
else if (avg >= 50)
    grade = "C";
else
    grade = "Fail";

document.getElementById("output").innerHTML =
    "Average Marks: " + avg + "<br>Grade: " + grade;
}
</script>
</body>
</html>
```

OUTPUT:

UIDCSE25242-SK.TASMIYA

Student Marks and Grade

Subject 1:

Subject 2:

Subject 3:

Subject 4:

Subject 5:

Subject 6:

Average Marks: 75.5
Grade: A

Explanation of tags:

<button>: Creates a clickable button to calculate average and grade.

onclick: Event attribute used to call the JavaScript function when the button is clicked.

<div id="output">: Used to display the calculated average and grade on the screen.

<script>: Used to include JavaScript code inside the HTML file.

function: Defines a block of code that performs a specific task (calculating average and grade).

getElementById(): JavaScript method used to access HTML elements using their ID.

value: Retrieves the values entered in the input fields.

parseInt(): Converts the input values into integers.

if-else: Conditional statements used to assign grades based on the average marks.

innerHTML: Used to display the result inside the <div> element.

WEEK-11

AIM:

A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters.

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Username Validation</title>
</head>
<body>
<label>Enter Username:</label>
<input type="text" id="username" oninput="checkUsername()">
<p id="message"></p>
<script>
function checkUsername() {
  let username = document.getElementById("username").value;
  let message = document.getElementById("message");
  if (username.length < 5) {
    message.innerHTML = "Username must be at least 5 characters";
    message.style.color = "red";
  } else {
    message.innerHTML = "Valid username";
    message.style.color = "green";
  }
}
</script>
</body>
</html>
```

OUTPUT:



Explanation of tags:

<head>: The element contains meta information about the HTML page.

<title>: The <title> element specifies a title for the HTML page

<body>: The <body> element contains all visible content like text, images, links, tables, etc.

<script>: Used to include JavaScript code inside the HTML file.

getElementById(): JavaScript method used to access HTML elements using their ID.

innerHTML: to display the result inside the <div> element.

B).AIM:

A registration form should validate email format before submission. If invalid, the form should not be submitted.

CODE:

```
<!DOCTYPE html>

<html>

<head>

  <title>Email Validation</title>

</head>

<body>

<form onsubmit="return validateEmail()">

  <label>Enter Email:</label>

  <input type="text" id="email">

  <p id="message"></p>

  <button type="submit">Register</button>

</form>

<script>
```



```

function validateEmail() {
    let email = document.getElementById("email").value;
    let message = document.getElementById("message");
    let pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!pattern.test(email)) {
        message.innerHTML = "Invalid email format";
        message.style.color = "red";
        return false;
    }
    return true;
}
</script>
</body>
</html>

```

OUTPUT:



Explanation of tags:

<head>: The element contains meta information about the HTML page.

<title>: The <title> element specifies a title for the HTML page

<body>: The <body> element contains all visible content like text, images, links, tables, etc.

<script>: Used to include JavaScript code inside the HTML file.

getElementById(): JavaScript method used to access HTML elements using their ID.

innerHTML: to display the result inside the <div> element.

C).AIM:

A system should check password strength while typing and display whether it is weak or strong.

CODE:

```
<!DOCTYPE html>

<html>

<head>

  <title>Password Strength</title>

</head>

<body>

  <label>Enter Password:</label>

  <input type="password" id="password" oninput="checkPassword()">

  <p id="message"></p>

  <script>

function checkPassword() {

  let password = document.getElementById("password").value;

  let message = document.getElementById("message");

  let strongPattern = /^(?=.*[A-Z])(?=.*[a-z])(?=.*\d){8,}$/;

  if (password.length === 0) {

    message.innerHTML = "";

  }

  else if (strongPattern.test(password)) {

    message.innerHTML = "Strong Password";

    message.style.color = "green";

  }

  else {

    message.innerHTML = "Weak Password";

    message.style.color = "red";

  }

}

</script>
```

```
</body>
```

```
</html>
```

OUTPUT:



Explanation of tags:

<head>: The element contains meta information about the HTML page.

<title>: The <title> element specifies a title for the HTML page

<body>: The <body> element contains all visible content like text ,images ,links ,tables ,etc.

<script>: Used to include JavaScript code inside the HTML file.

getElementById(): JavaScript method used to access HTML elements using their ID.

innerHTML: to display the result inside the <div> element

D).AIM:

A form should validate a mobile number when the user leaves the input field. The number must be exactly 10 digits.

CODE:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Mobile Validation</title>
```

```
</head>
```

```
<body>
```

```
<label>Enter Mobile Number:</label>
```

```
<input type="text" id="mobile" onblur="validateMobile()">
```

```
<p id="message"></p>
<script>
function validateMobile() {
    let mobile = document.getElementById("mobile").value;
    let message = document.getElementById("message");

    let pattern = /^[0-9]{10}$/;

    if (pattern.test(mobile)) {
        message.innerHTML = "Valid Mobile Number";
        message.style.color = "green";
    } else {
        message.innerHTML = "Mobile number must be exactly 10 digits";
        message.style.color = "red";
    }
}
</script>
</body>
</html>
```

OUTPUT:



Explanation of tags:

<head>: The element contains meta information about the HTML page.

<title>: The <title> element specifies a title for the HTML page

<body>: The <body> element contains all visible content like text ,images ,links ,tables ,etc.

<script>: Used to include JavaScript code inside the HTML file.

getElementById(): JavaScript method used to access HTML elements using their ID.

innerHTML: to display the result inside the <div> element.

E).AIM:

Design an HTML page, such that the registration form must validate

CODE:

```
<!DOCTYPE html>

<html>

<head>

  <title>Registration Form</title>

</head>

<body>

<form onsubmit="return validateForm()">

  <!-- Username -->

  <label>Username:</label>

  <input type="text" id="username" oninput="checkUsername()">

  <p id="userMsg"></p>

  <!-- Email -->

  <label>Email:</label>

  <input type="text" id="email">

  <p id="emailMsg"></p>

  <!-- Password -->

  <label>Password:</label>

  <input type="password" id="password" oninput="checkPassword()">

  <p id="passMsg"></p>

  <!-- Mobile -->

  <label>Mobile:</label>

  <input type="text" id="mobile" onblur="checkMobile()">

  <p id="mobileMsg"></p>

  <button type="submit">Register</button>
```

```
</form>

<script>
function checkUsername() {
    let u = document.getElementById("username").value;
    let msg = document.getElementById("userMsg");
    if (u.length < 5) {
        msg.innerHTML = "Minimum 5 characters required";
        msg.style.color = "red";
    } else {
        msg.innerHTML = "";
    }
}

function checkPassword() {
    let p = document.getElementById("password").value;
    let msg = document.getElementById("passMsg");
    let pattern = /^(?=.*[A-Z])(?=.*[a-z])(?=.*\d){8}$/;
    if (pattern.test(p)) {
        msg.innerHTML = "Strong Password";
        msg.style.color = "green";
    } else {
        msg.innerHTML = "Weak Password";
        msg.style.color = "red";
    }
}

function checkMobile() {
    let m = document.getElementById("mobile").value;
    let msg = document.getElementById("mobileMsg");
    let pattern = /^[0-9]{10}$/;
    if (pattern.test(m)) {
        msg.innerHTML = "";
    }
}
```

```
} else {  
    msg.innerHTML = "Enter exactly 10 digits";  
    msg.style.color = "red";  
}  
}  
function validateForm() {  
    let email = document.getElementById("email").value;  
    let emailMsg = document.getElementById("emailMsg");  
  
    let pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;  
  
    if (!pattern.test(email)) {  
        emailMsg.innerHTML = "Invalid Email";  
        emailMsg.style.color = "red";  
        return false;  
    }  
    return true;  
}  
</script>  
</body>  
</html>
```

OUTPUT:



Registration Form

Username:

Email:

Password:

Strong Password

Mobile:

Explanation of tags:

<head>:The element contains meta information about the HTML page.

<title>:The <title> element specifies a title for the HTML page

<body>:The <body>element contains all visible content like text ,images ,links ,tables ,etc.

<script>: Used to include JavaScript code inside the HTML file.

getElementById(): JavaScript method used to access HTML elements using their ID.

innerHTML: to display the result inside the <div> element.

WEEK-12

AIM:

Explain the setup process of electron application in your desktop .

Set up process of electron application :

- a) At first we need to download the latest released version of node.js from the web browser .
- b) Later on set up the node.js in the system.
- c) Now create a folder on the desktop ,for easy accessibility .
- d) Now open windows power shell as administrator give command "Set-Execution Policy Remote Signed" after giving that it asks for N or Y ,now press "Y" .
- e) Later on open visual studio code and open the folder created by you at starting ,now create a file in the folder and name it as main.js .
- f) Now open a new terminal and give prompt as "npm init -y" now all the packages will get downloaded.
- g) After that give prompt as "npm install electron --save-dev" now the electron application will get downloaded in your system .

OUTPUT:

